第三章 扭转

Chapter 3 Torsion

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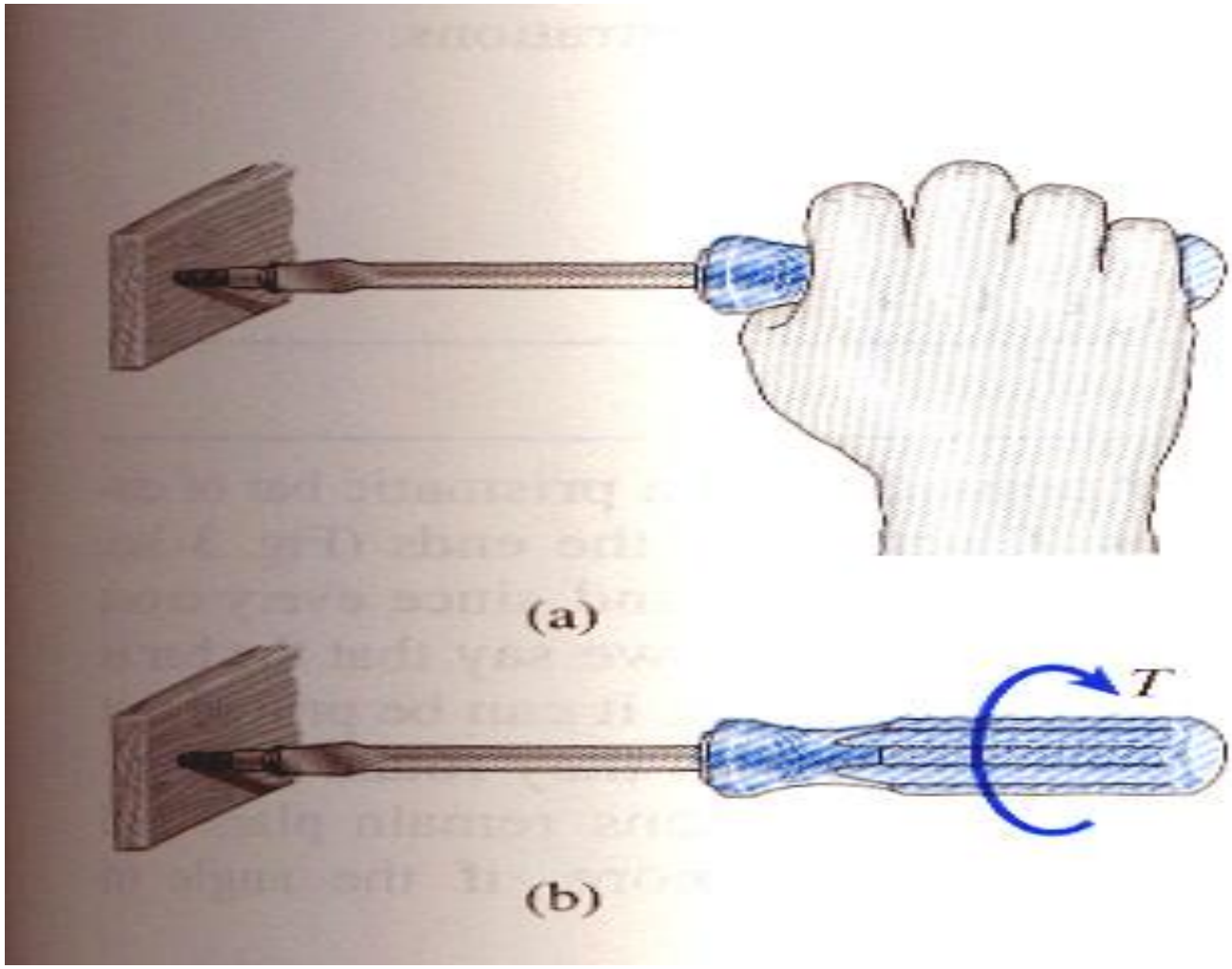
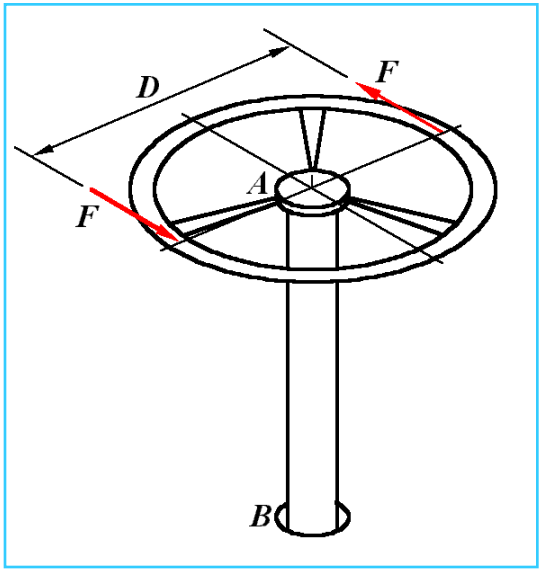
§ 3-7 非圆截面杆的扭转

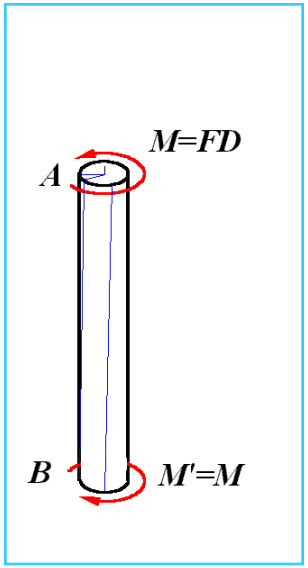
§ 3-1 扭转的概念及实例

(Concepts and example problem of torsion)

一、工程实例 (Example problems)




D
F
A
F
B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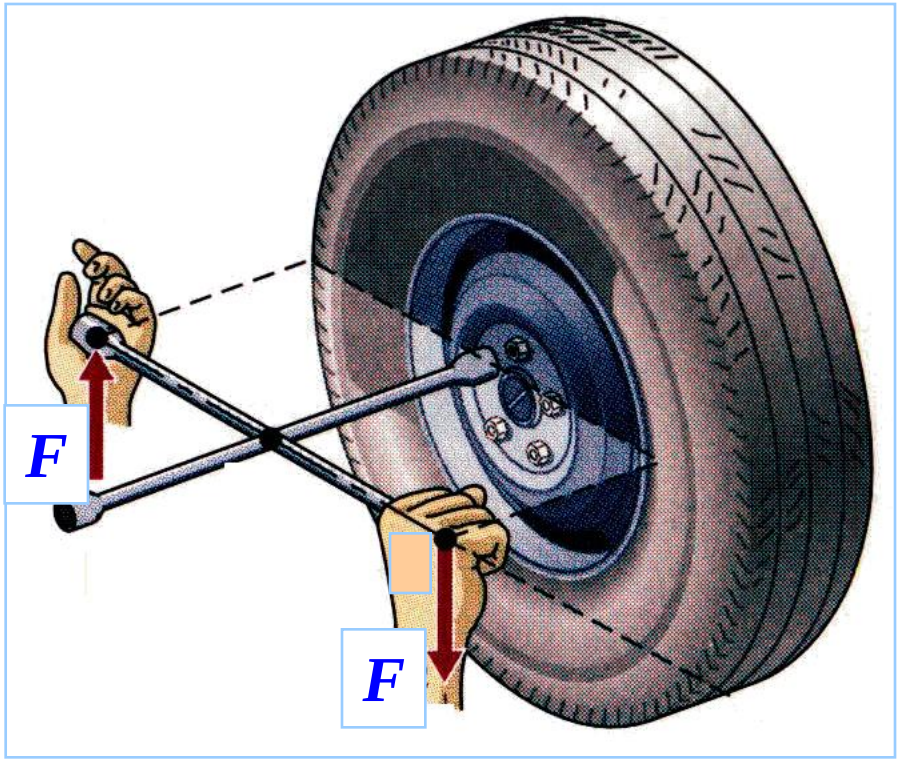
A
B
M'=M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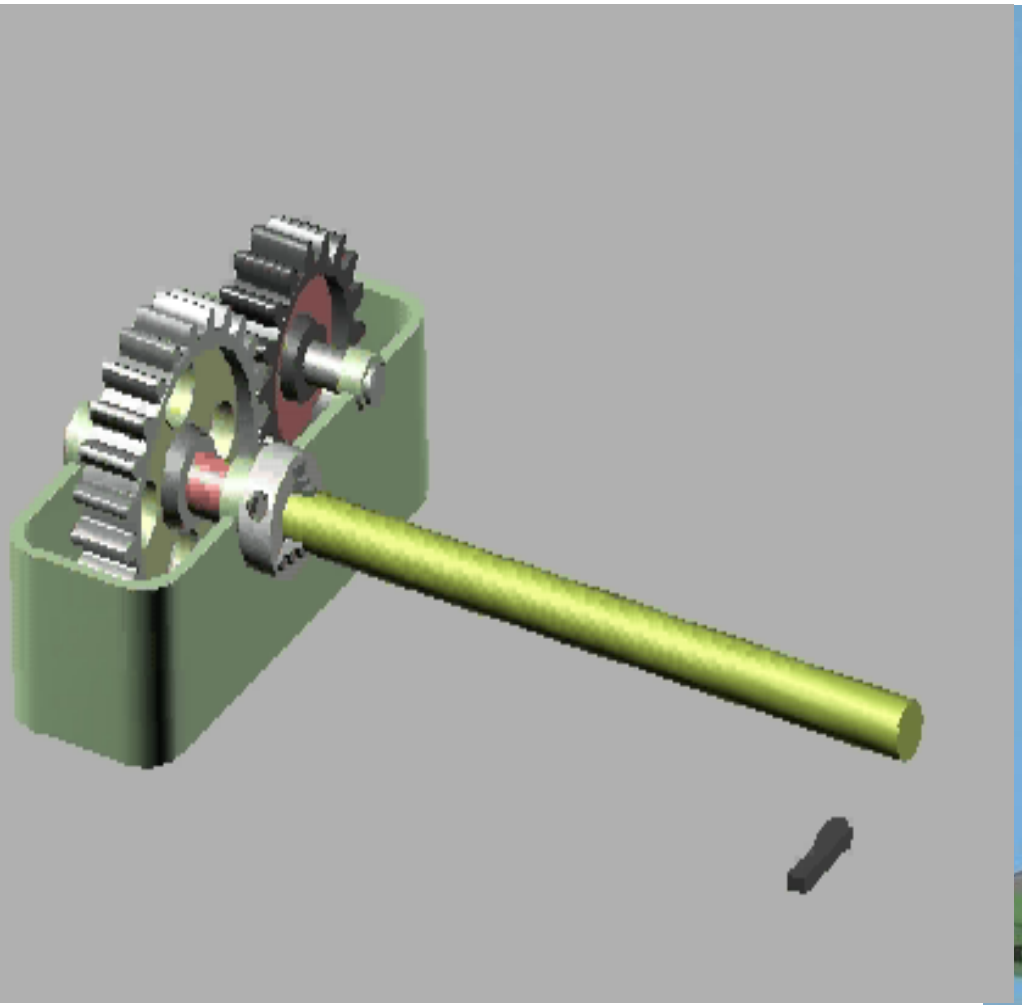
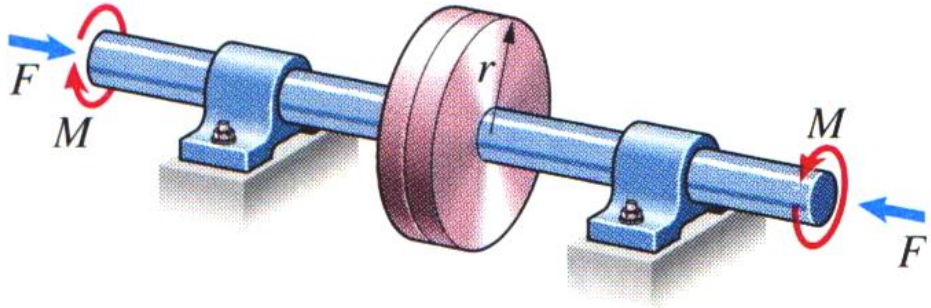


Tc-1600


F
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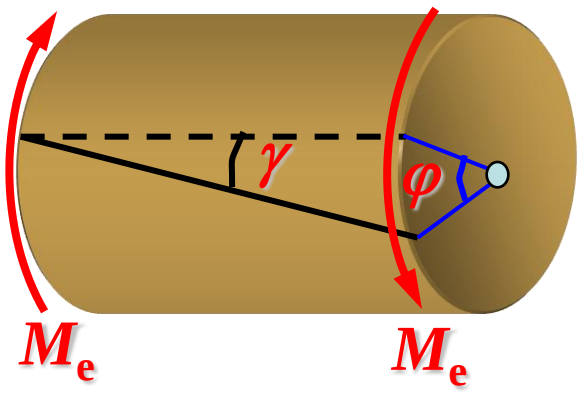


二、受力特点 (Character of external force)

杆件的两端作用两个大小相等、方向相反、且作用平面垂直于杆件轴线的力偶。

三、变形特点 (Character of deformation)

杆件的任意两个横截面都发生绕轴线的相对转动。

四、研究对象

轴－以扭转为主要变形的杆件。

主要研究圆形或圆环形等直杆的的扭转，
对于非圆截面只了解。

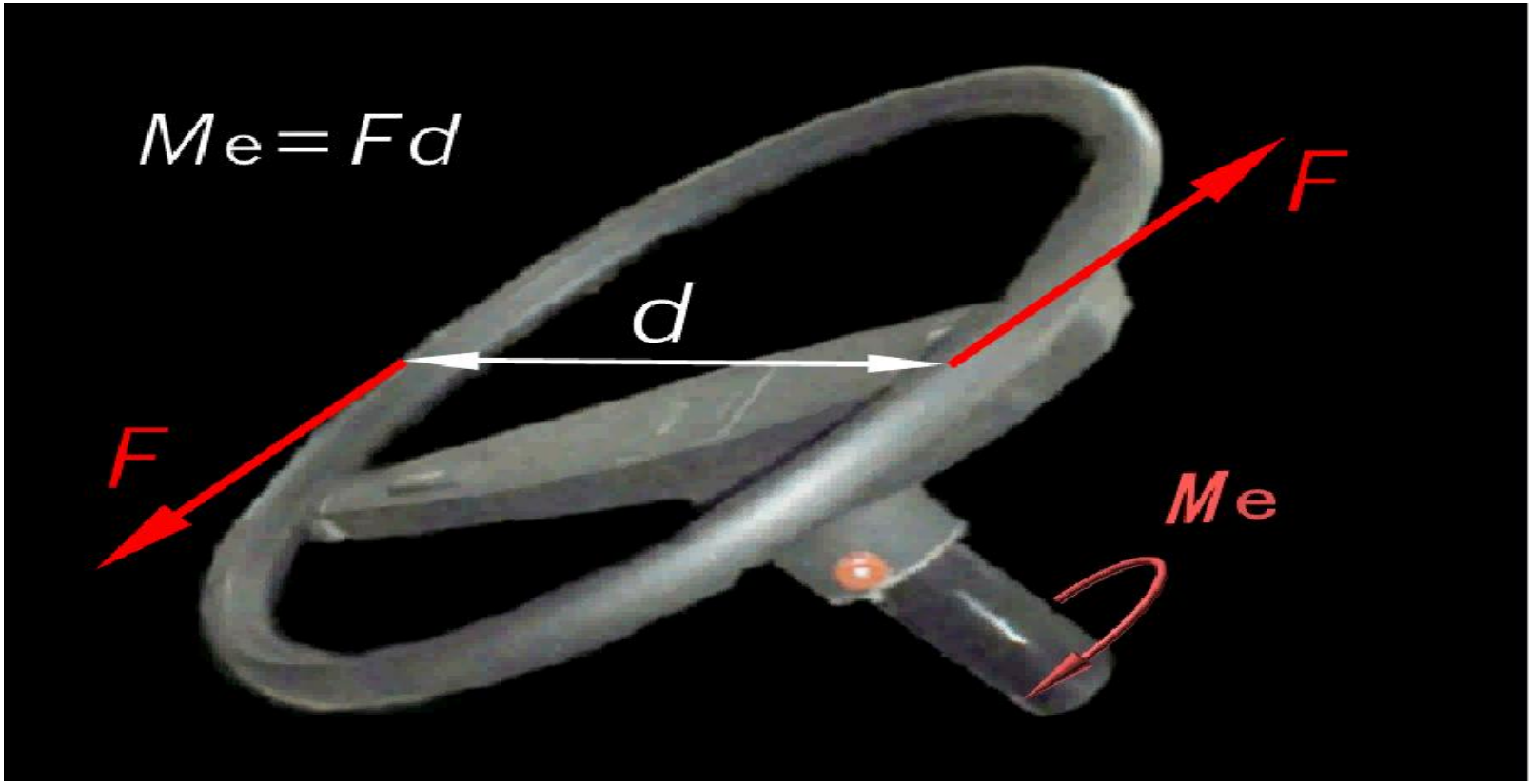
机器中的传动轴、石油钻机中的钻杆等。

§3-2 扭转的内力的计算

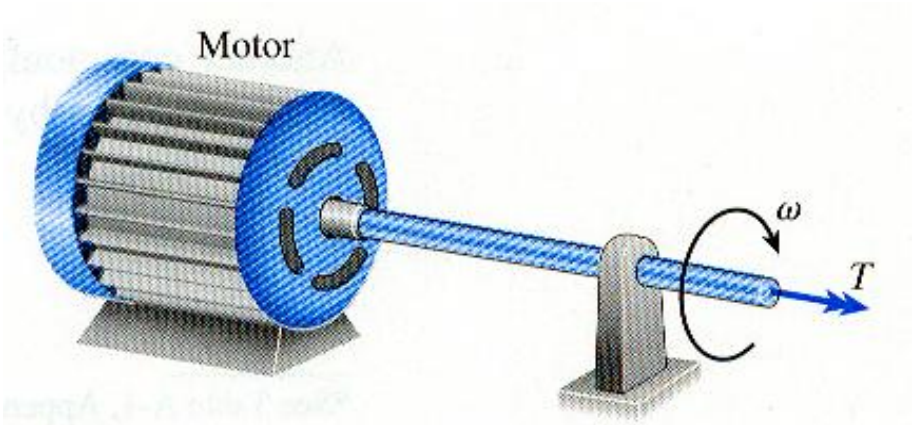
(Calculating internal force of torsion)

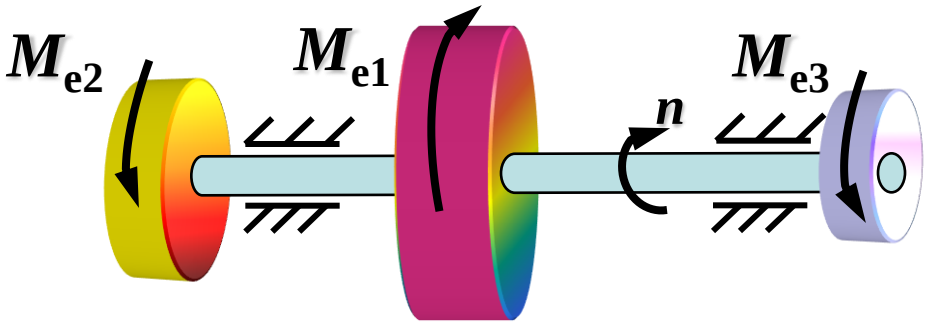
一、外力偶矩的计算 (Calculation of external moment)

1、直接计算



2、按输入功率和转速计算





n—轴的转速(r/min)　P—轴传递的功率(kW)

Me—作用在轴上的力偶矩(N ·m)

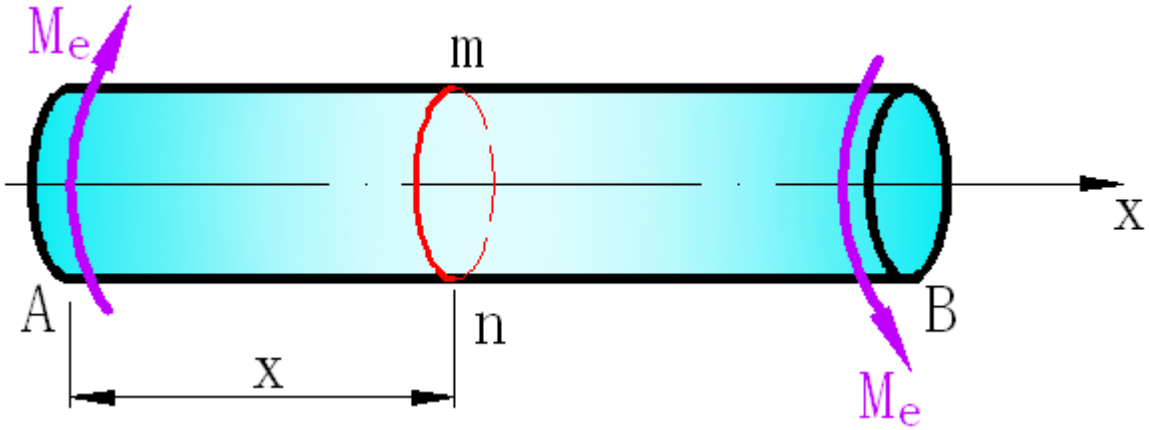
$$P = m_e \cdot \omega \quad \Longrightarrow \quad P \times 10^3 = m_e \cdot \frac{n\pi}{30}$$

$$\{m_e\}_{N\cdot m} = 9549 \cdot \frac{\{P\}_{kW}}{\{n\}_{r/min}}$$

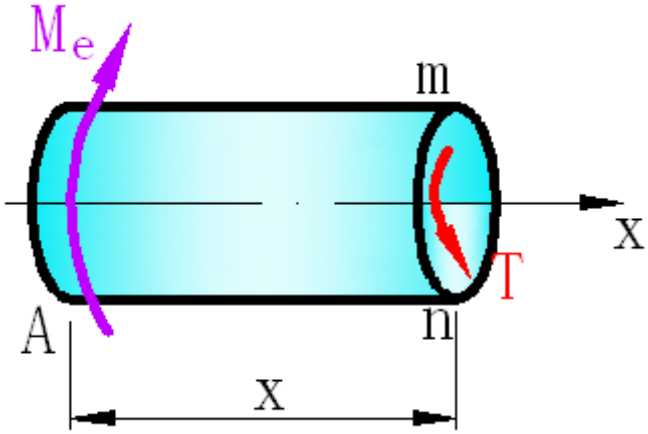
$$\{m_e\}_{N\cdot m} = 7024 \cdot \frac{\{P\}_{马力}}{\{n\}_{r/min}}$$

二、内力的计算 (Calculation of internal force)

1、扭矩(Calculating internal force)

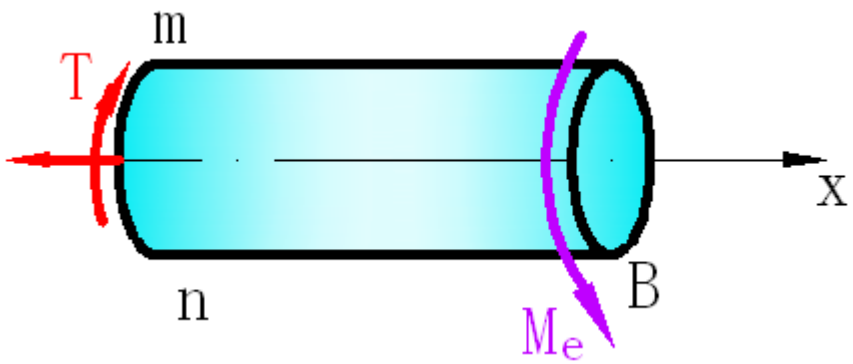


截面法



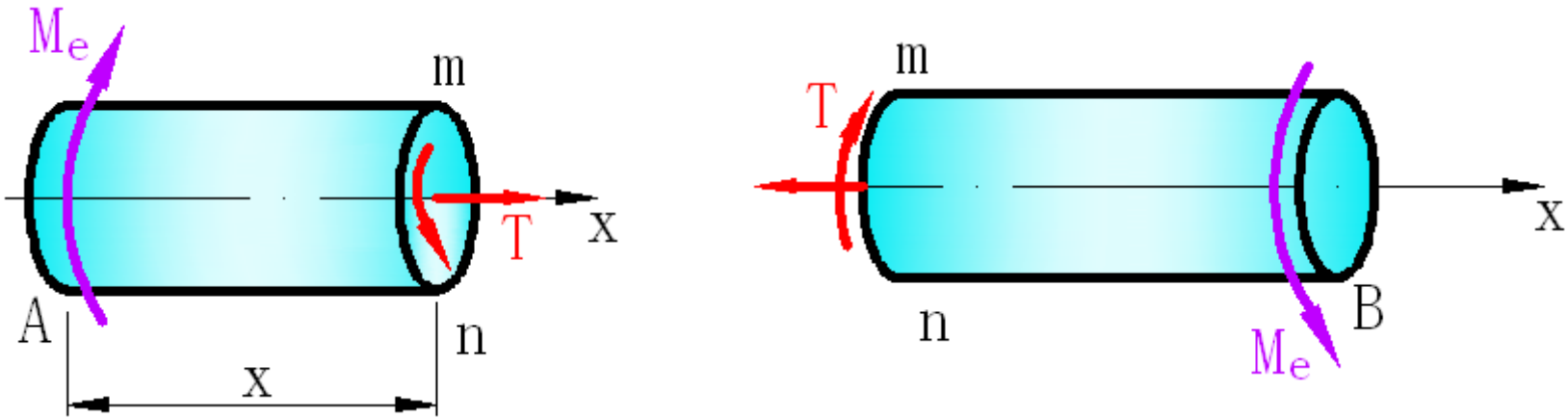
$T=M_e$

T与x无关。



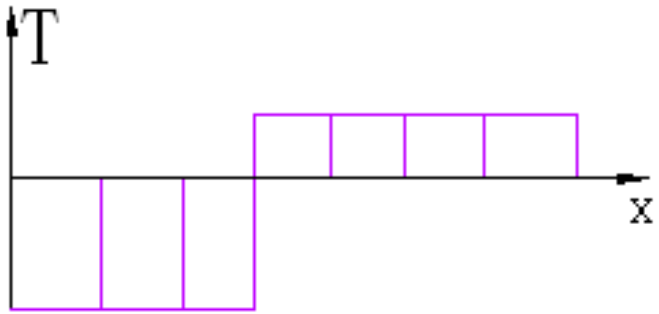
2、扭矩符号的规定(Sign convention for torque)

采用右手螺旋法则，当力偶矩矢的指向背离截面时扭矩为正，反之为负。

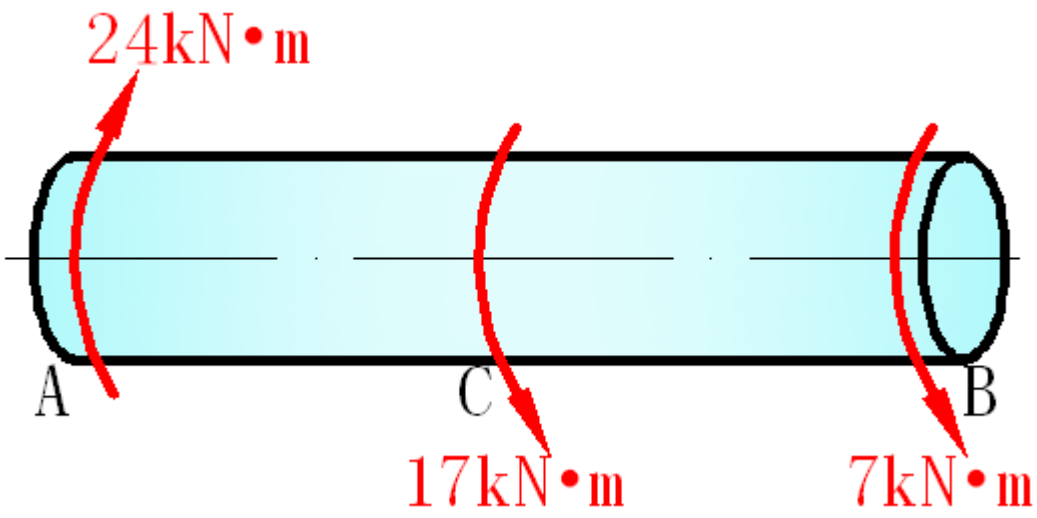


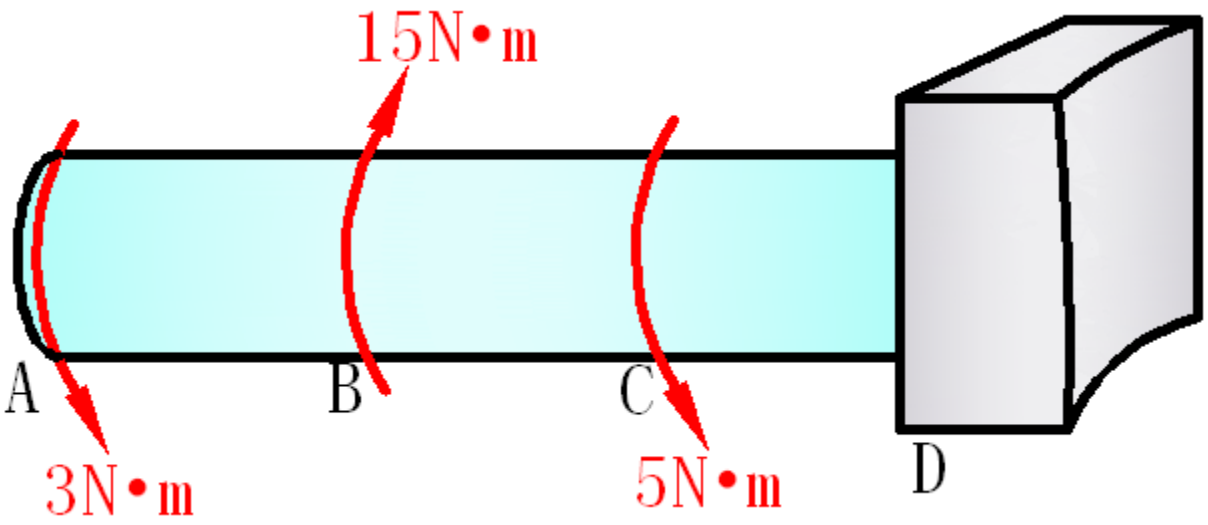
3、扭矩图(Torque diagram)

用平行于杆轴线的坐标x表示横截面的位置，用垂直于杆轴线的坐标T 示横截面上的扭矩，正的扭矩画在 x 轴上方,负的扭矩画在x轴下方。

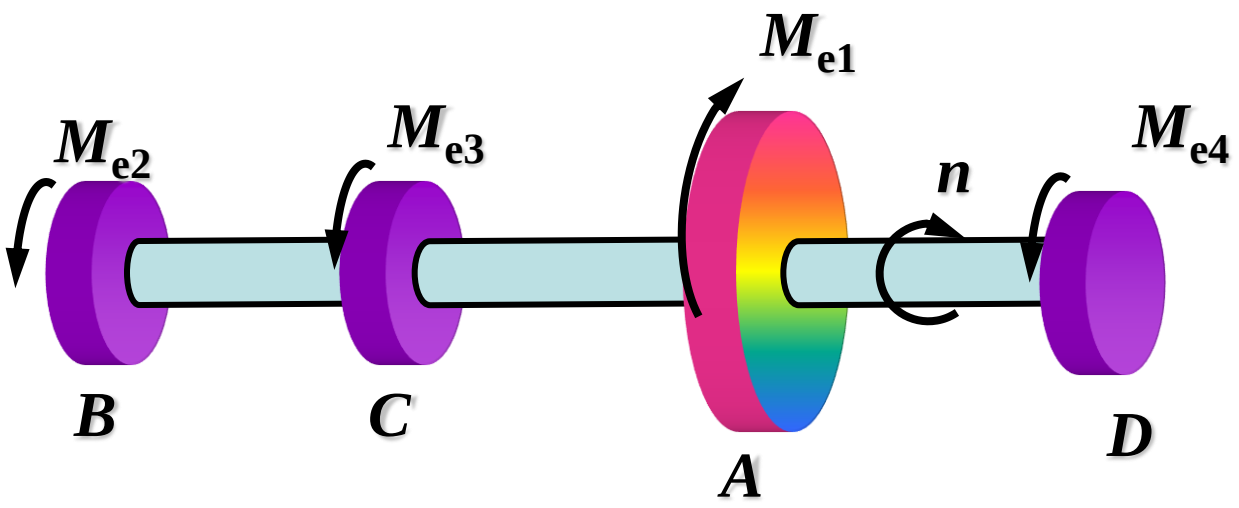


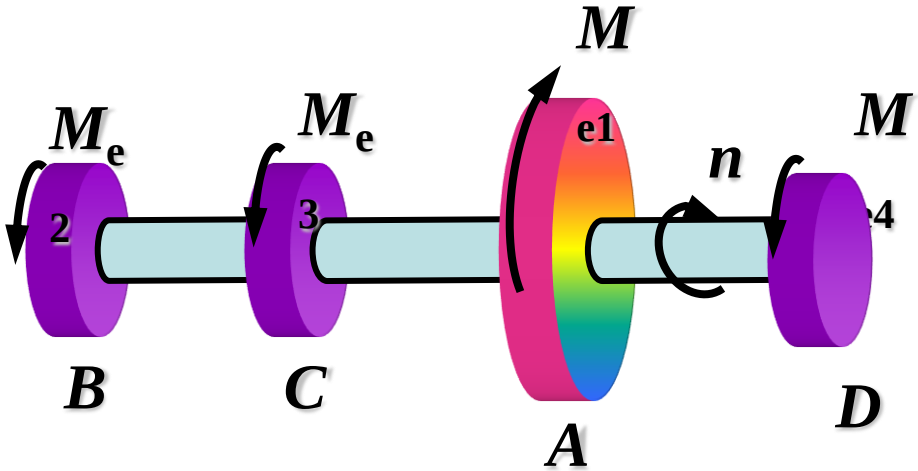
例题3-1　作扭矩图。

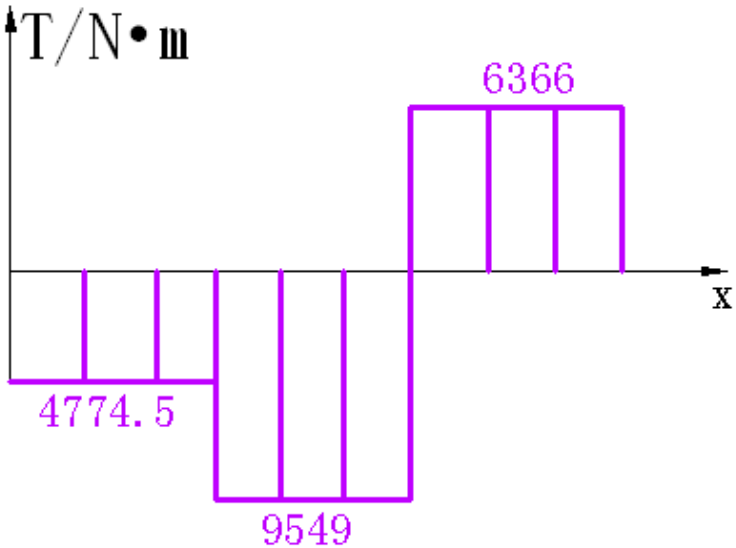


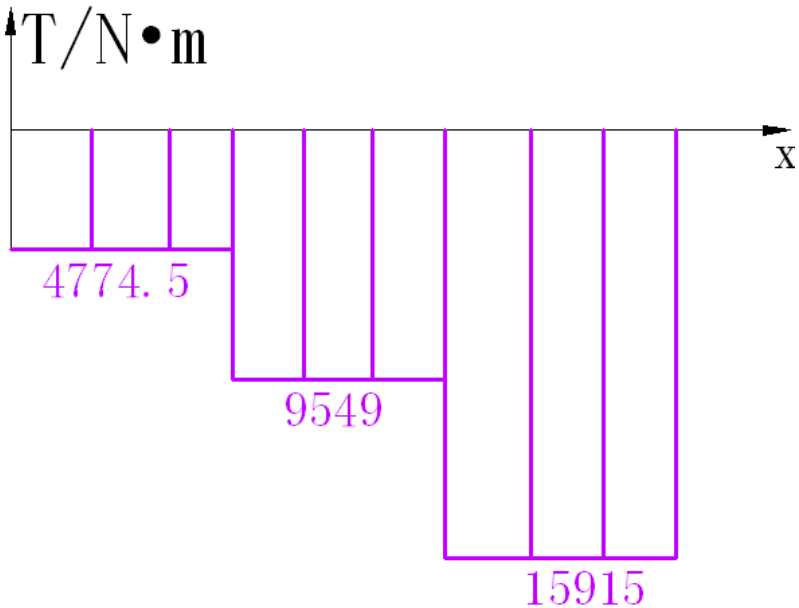


例题3-2　一传动轴如图所示,其转速 n = 300 r/min，主动轮A输入的功率为P_1 = 500 kW。三个从动轮输出的功率分别为P_2 = 150 kW，P_3 = 150 kW，P_4 = 200 kW。若不计轴承摩擦所耗的功率。试做扭矩图。









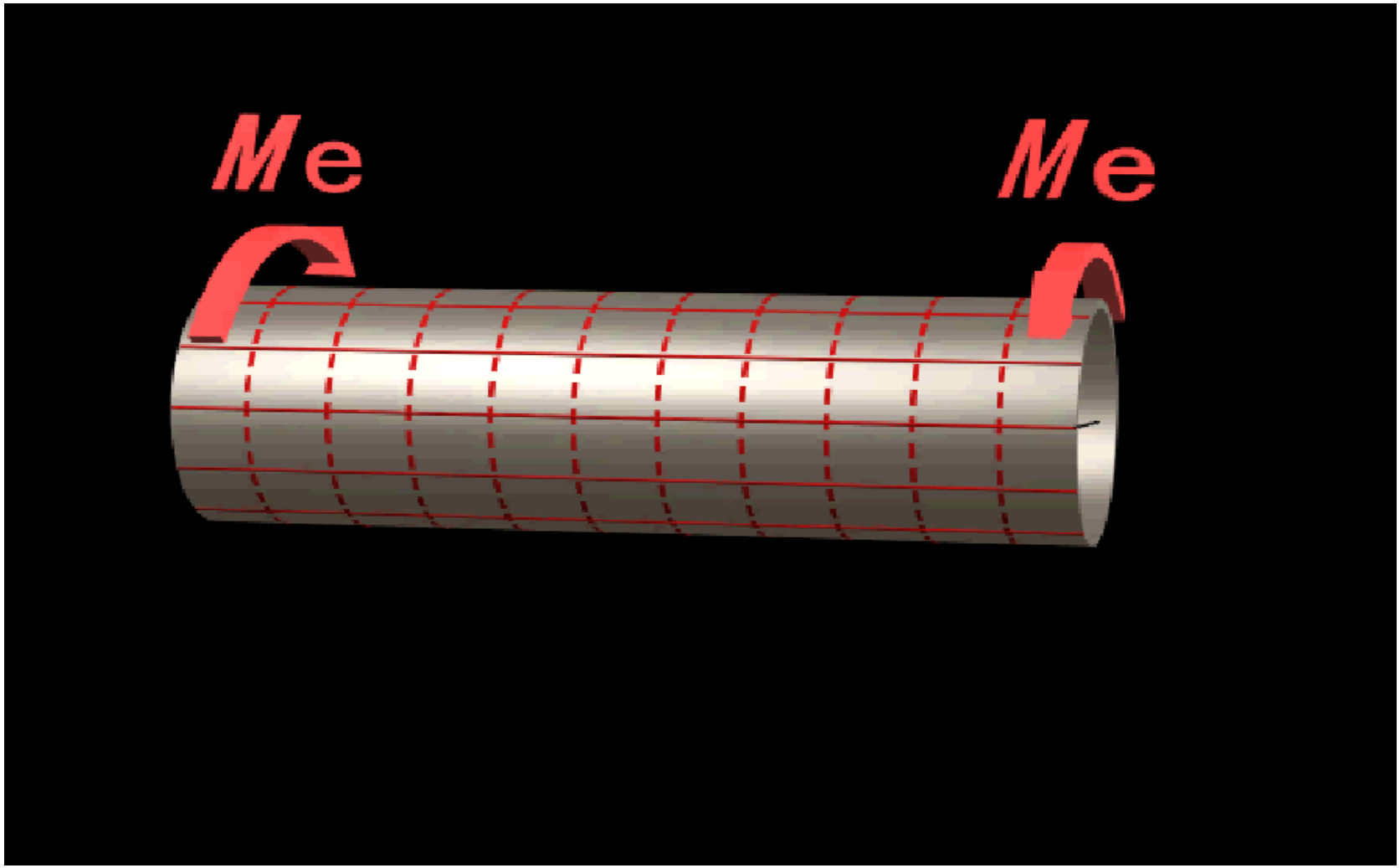
$$M_{e1} = 15915 \ \ \mathrm{N \cdot m}$$

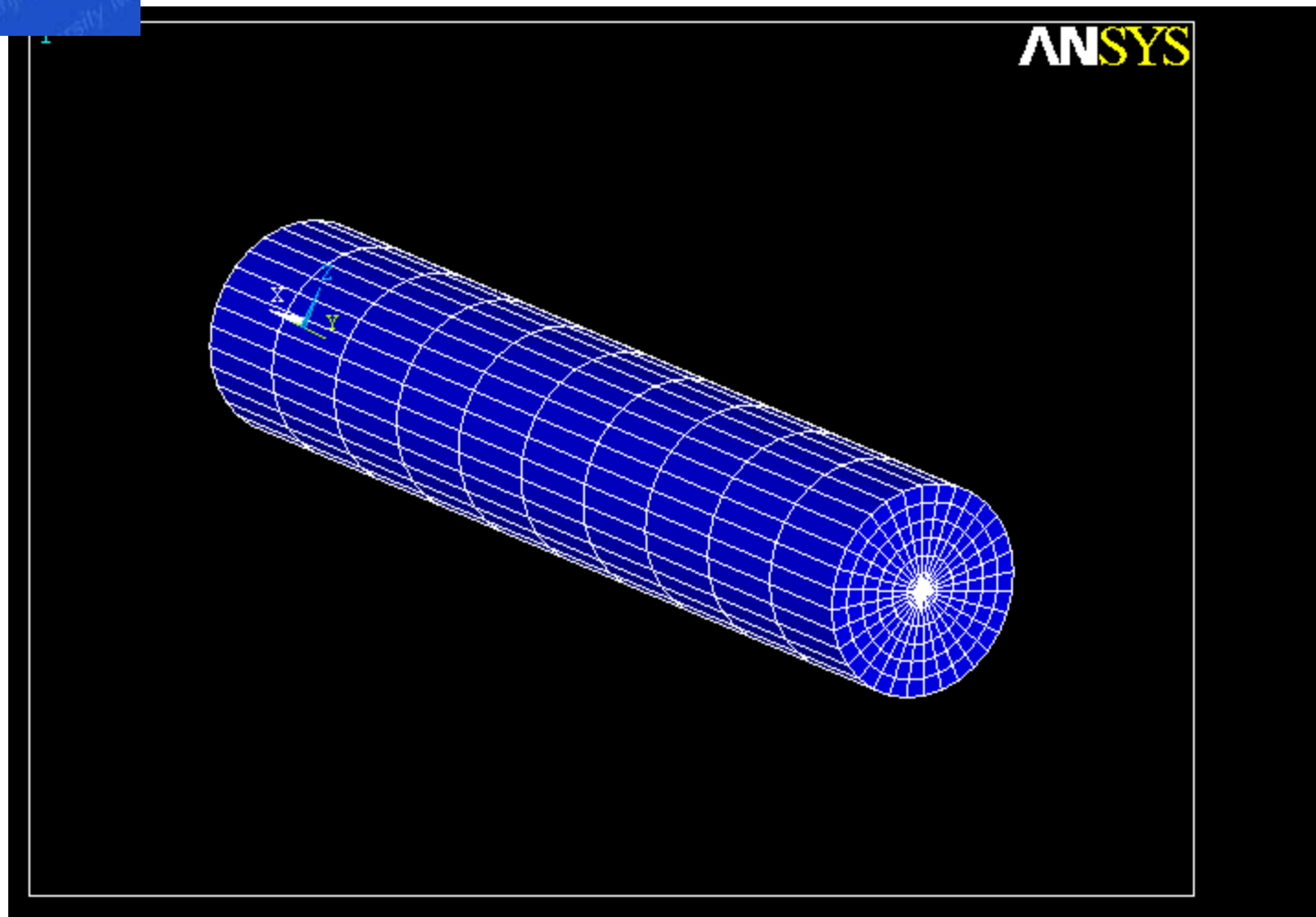
$$M_{e2} = M_{e3} = 4774.5 \ \ \mathrm{N \cdot m}$$

$$M_{e4} = 6366 \ \ \mathrm{N \cdot m}$$

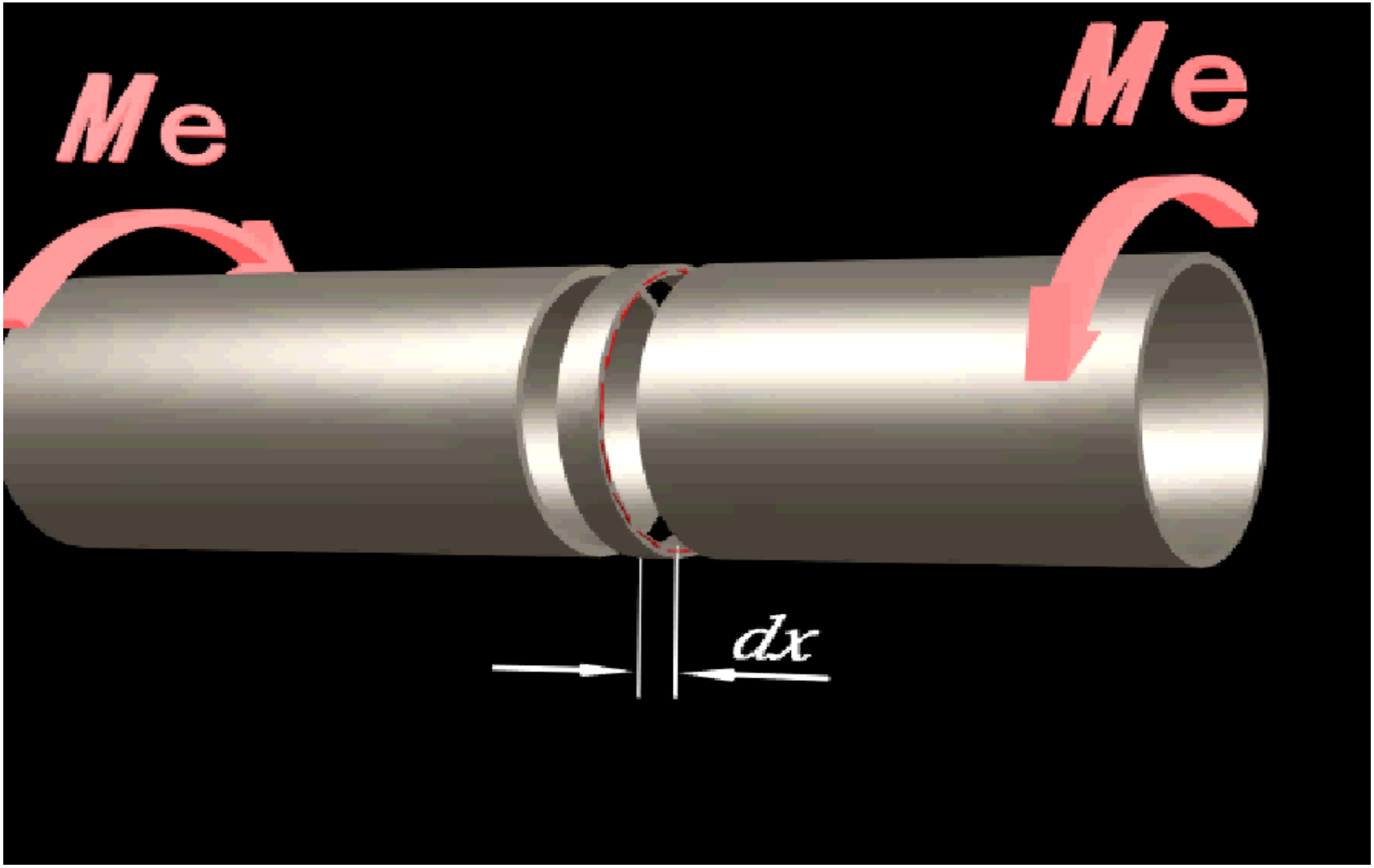
§3-2 纯剪切（pure shear）

圆　筒　变　形

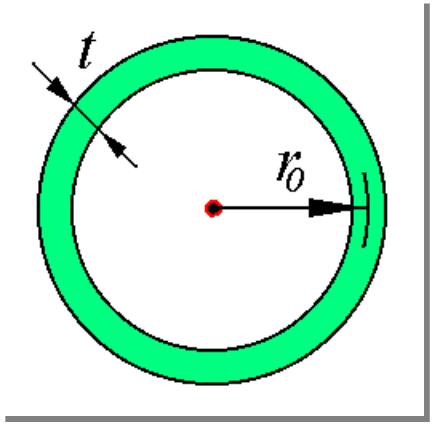



ANSYS
X
Y

圆 筒 单 元 体



薄壁圆筒：壁厚 $t \le \frac{1}{10} r_0$（r_0：为平均半径）

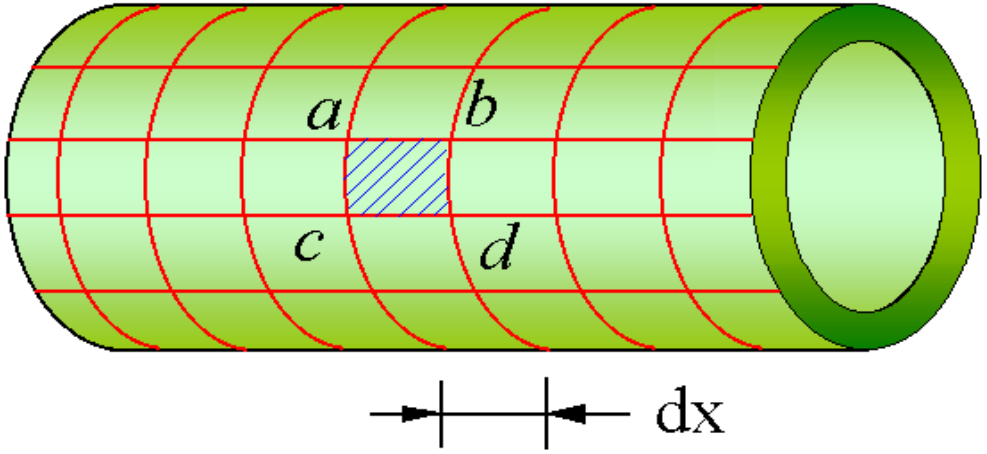


一、实验

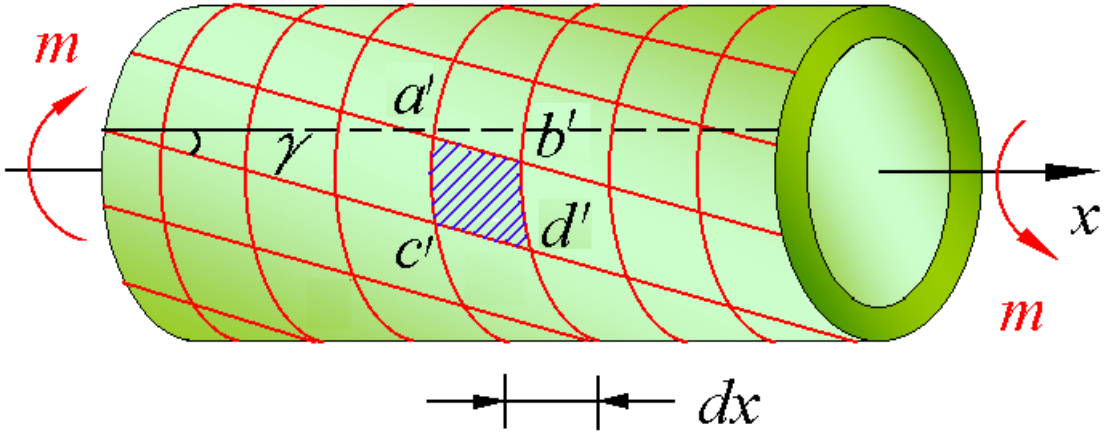
1、实验前

①绘纵向线，圆周线；

②施加一对外力偶 m。



2、实验后



（1）圆筒表面的各圆周线的形状、大小和间距均未改变；

（2）各纵向线均倾斜了同一微小角度γ；

（3）所有矩形网格均歪斜成同样大小的平行四边形。

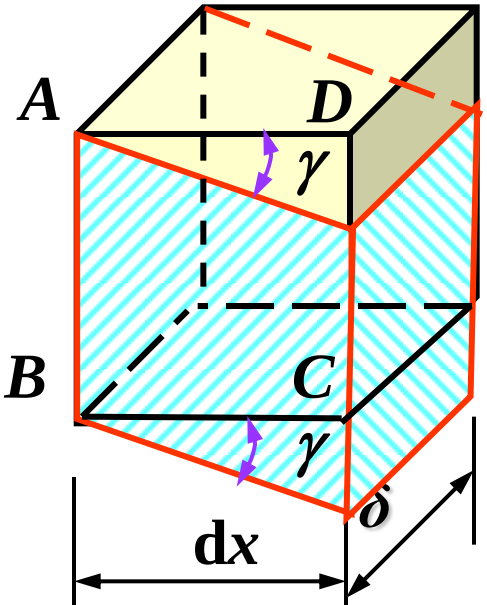
3、推论(Inference)

（1）横截面上无正应力，

只有切应力；

（2）切应力方向垂直半径

或与圆周相切。

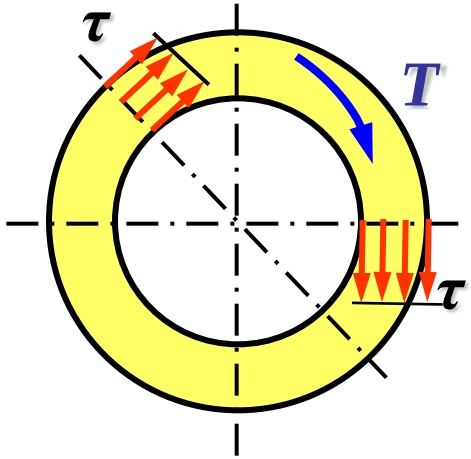


4、推导公式 (Derivation of formula)

圆周各点处切应力的方向于圆周相切，且数值相等，近似的认为沿壁厚方向各点处切应力的数值无变化。

$$\int_A \tau \mathrm{d}A \cdot r = \tau \cdot r \int_A \mathrm{d}A = \tau \cdot r(2\pi \cdot r \cdot \delta) = T$$

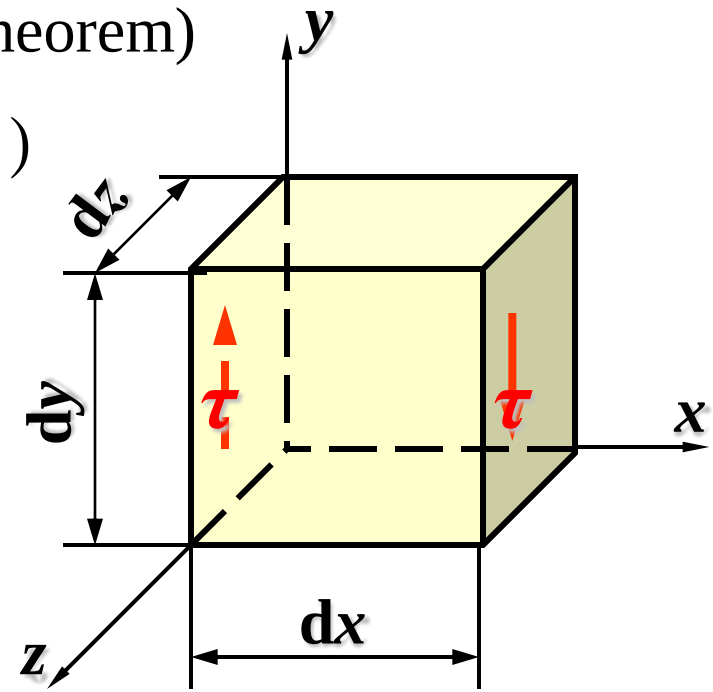
$$\tau = \frac{T}{2\pi r^2 \cdot \delta}$$



薄壁筒扭转时横截面上的切应力均匀分布，与半径垂直，指向与扭矩的转向一致，圆周各点处切应力的方向于圆周相切。

二、切应力互等定理 (Shearing Stress Theorem)

1、在单元体左、右面（杆的横截面）只有切应力，其方向于 y 轴平行，内力元素 $\tau dy\, dz$组成一个力偶。



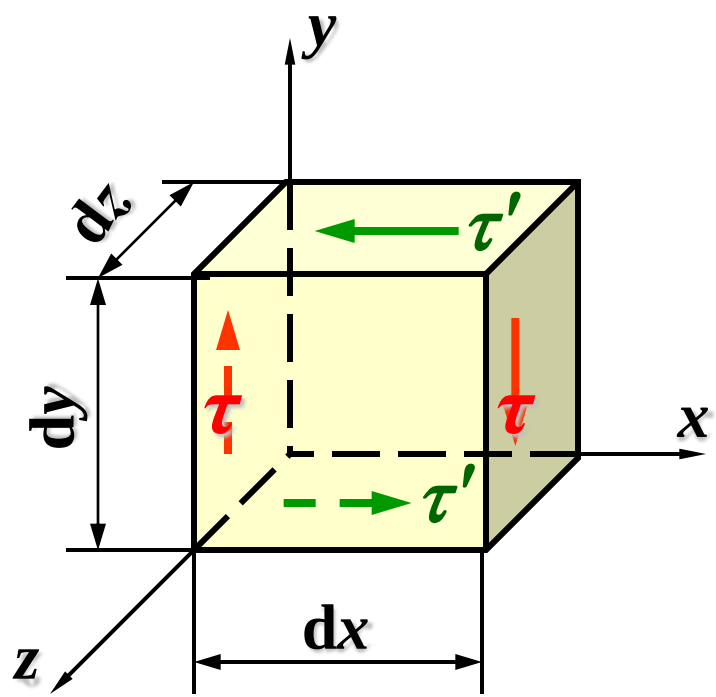
$\sum F_y=0$ $\quad M_z=(\tau dy\, dz)\, dx$。

$$\sum F_x=0 \qquad \sum M_y=0$$

$$\sum F_z=0 \qquad \sum M_x=0$$

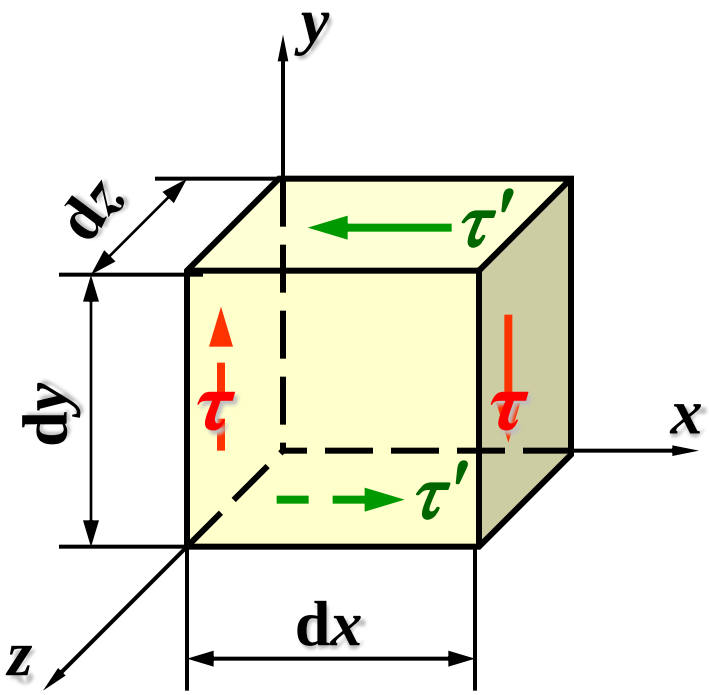
而 $\sum M_z=0$

在单元体的上、下两平面上必有大小相等，指向相反的一对内力元素它们组成力偶，其矩为 $(\tau' dy\, dz)\, dx$。



τdy dz) dx= (τ ´ dy dz) dx $\Longrightarrow$ $\tau' = \tau$

切应力互等定理：在单元体相互垂直的两个平面上，切应力必然成对出现，且数值相等，两者都垂直于两平面的交线，其方向则共同指向或共同背离该交线。



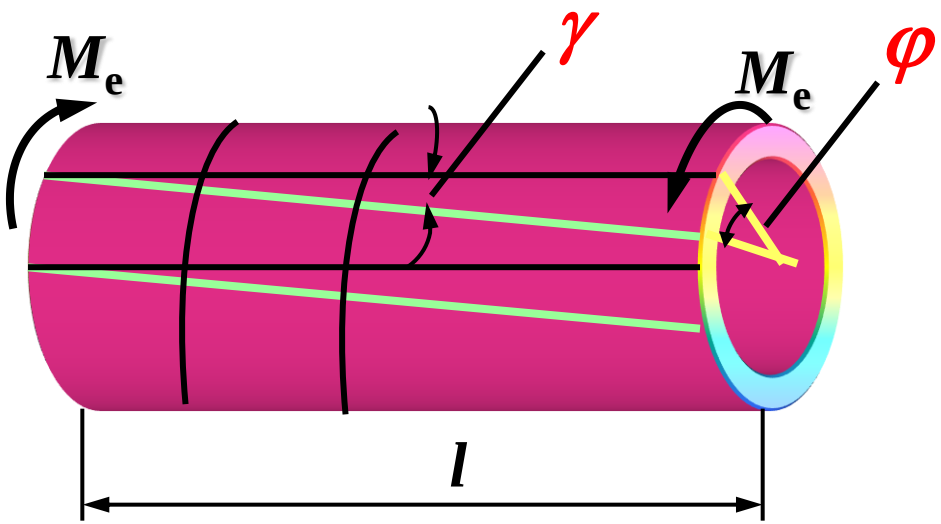
截面上存在正应力时，互等定理仍成立。

2、纯剪切单元体 (Element in pure shear)

单元体平面上只有切应力而无正应力，则称为纯剪切单元体。

三、剪切胡克定律(Hooke's law for shear)

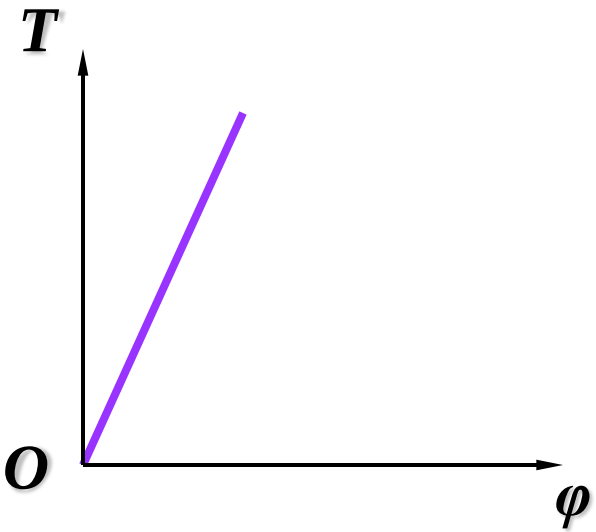
由图所示的几何关系得到

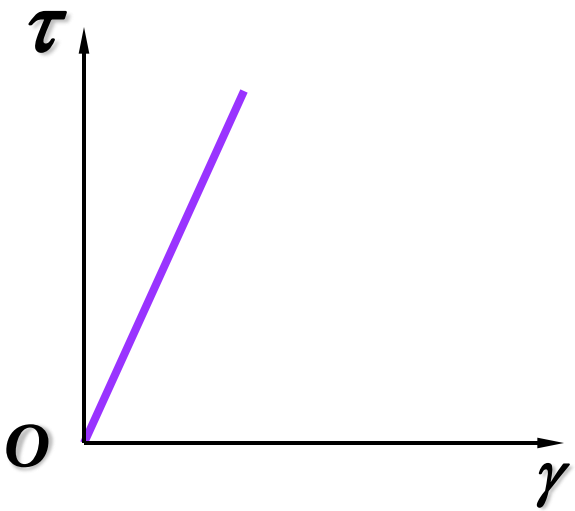


$$\gamma = \frac{r\varphi}{l}$$

式中，r 为薄壁圆筒的外半径。

$$\tau = \frac{T}{2\pi r^2 \delta}$$





从T与φ之间的线性关系,可推出τ与γ间的线性关系。

$$\boxed{\tau = G\gamma} \qquad \sigma = E\varepsilon$$

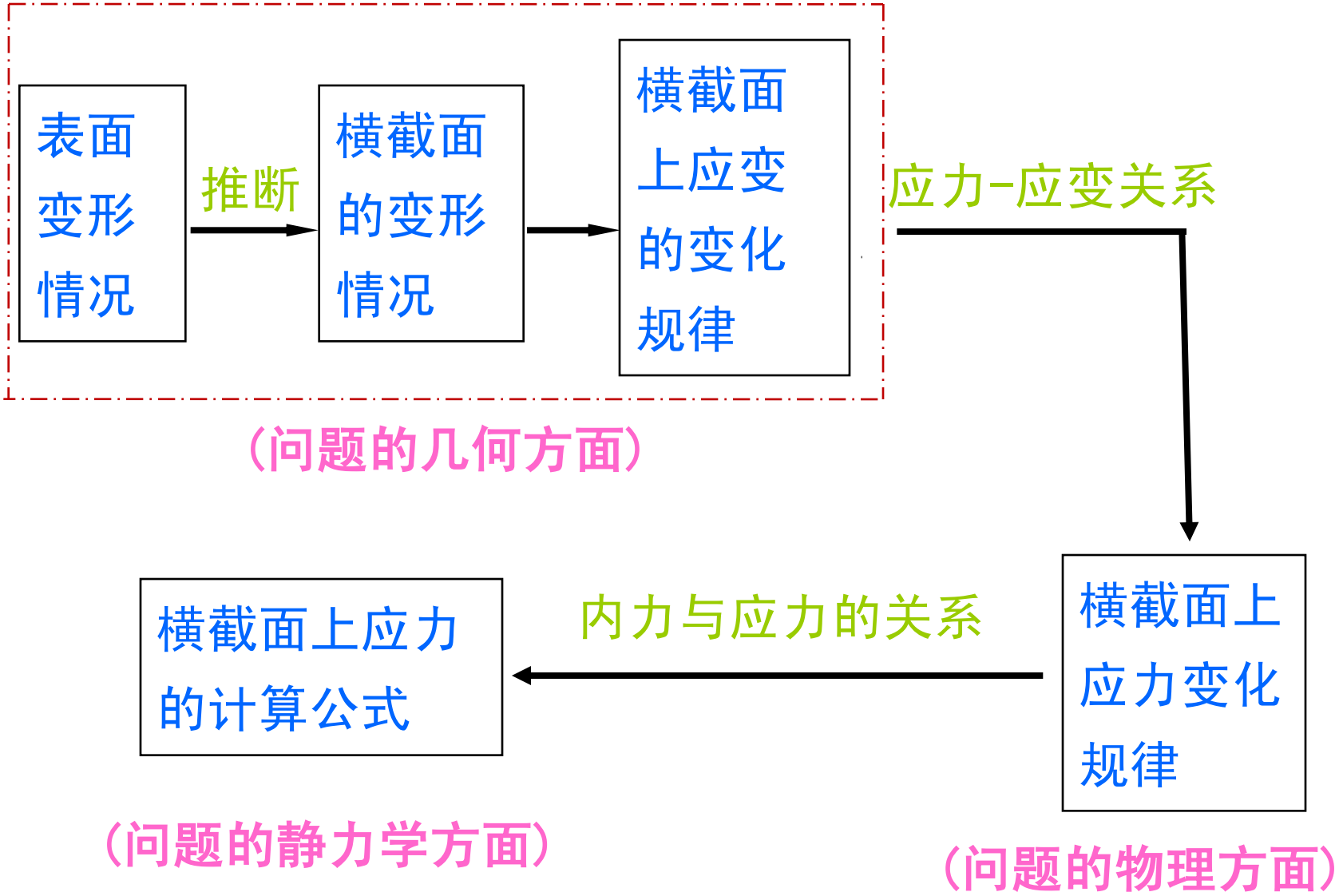
该式称为材料的剪切胡克定律。G是材料剪切弹性模量，因γ无量纲，故G的量纲与τ相同。不同材料的G值可通过实确定，钢材的G值约为80GPa。

剪切弹性模量、弹性模量和泊松比是表明材料弹性性质的三个常数。对各向同性材料，这三个弹性常数之间存在下列关系：

$$G = \frac{E}{2(1+\mu)}$$

在三个弹性常数中，只要知道任意两个，第三个量就可以推算出来。

§3-4 圆杆扭转时的应力（stress of circular bars）



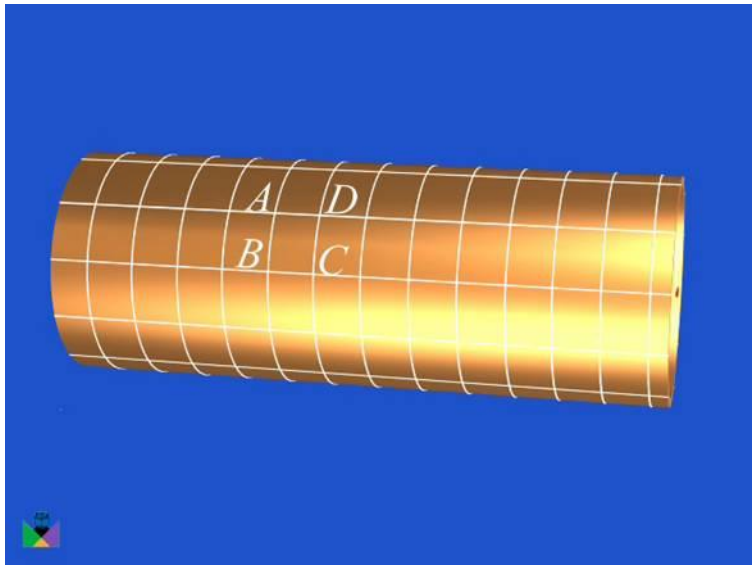
一、变形几何关系(Geometrical Relationship of Deformation)

1、变形现象(Deformation phenomenon)

（1）各圆周线的形状不变，仅绕轴线作相对转动。

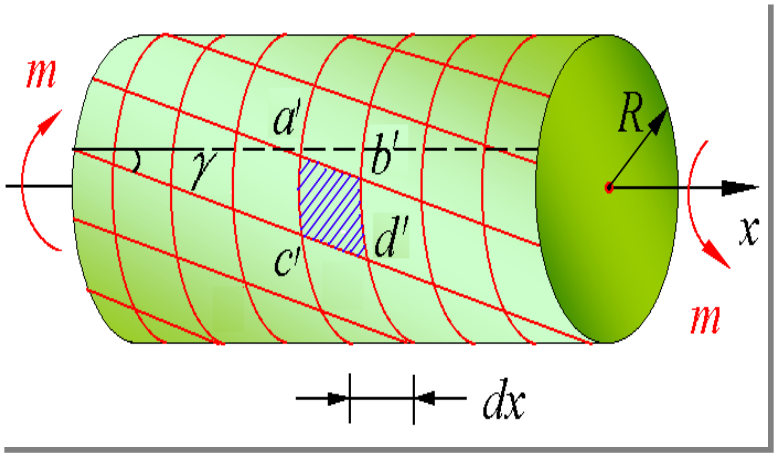
（2）当变形很小时，各圆周线的大小与间距均不改变。

（3）纵向线保持为直线，倾斜一个小角度。

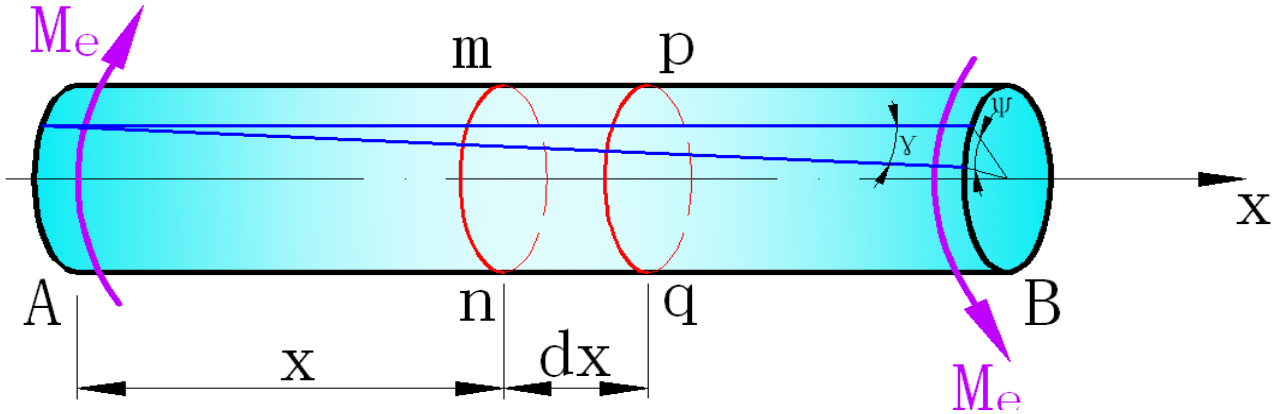


2、平面假设(Plane assumption)

各横截面如同刚性平面，仅绕轴线作相对转动。



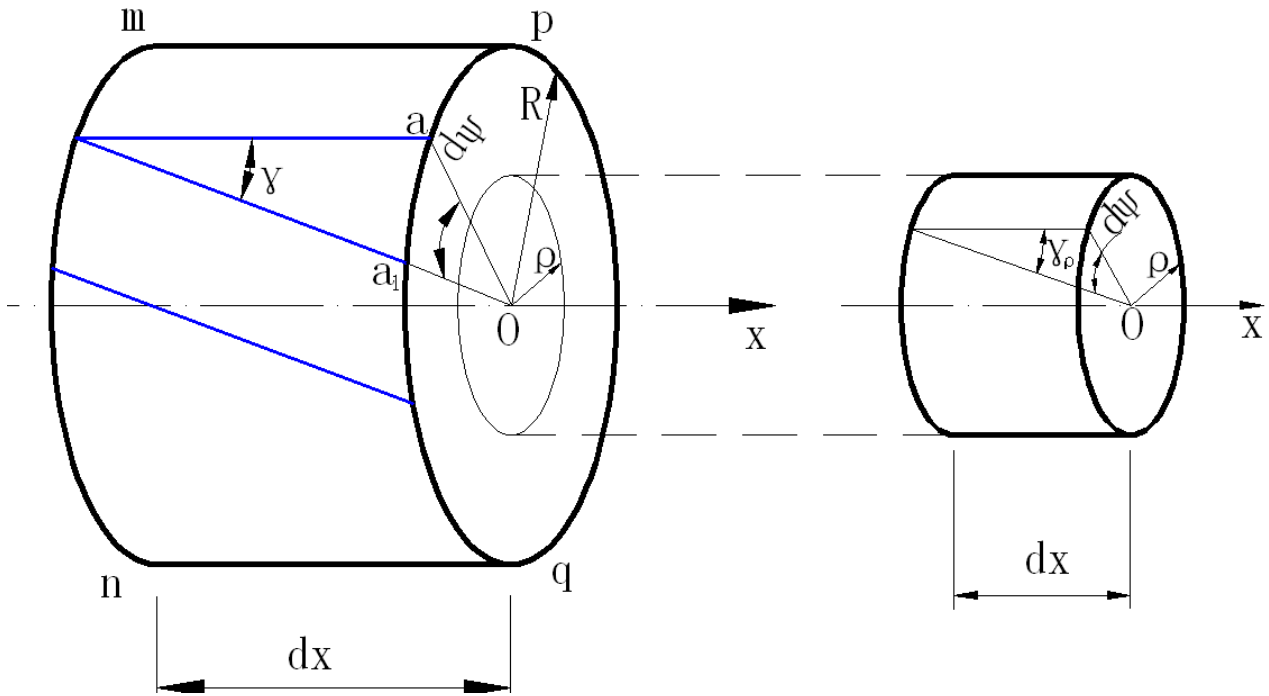
3、几何关系



$$\gamma = \frac{aa_1}{dx}$$

$$= \frac{Rd\varphi}{dx}$$

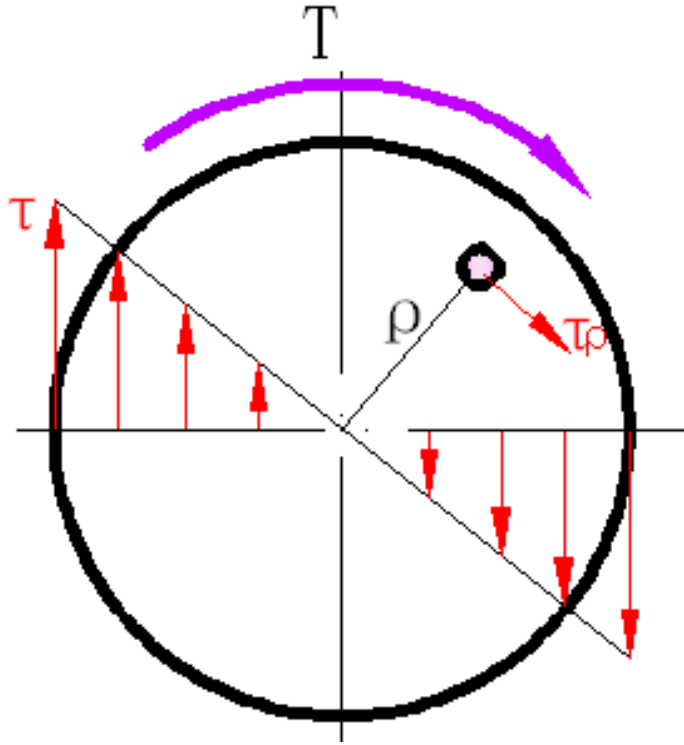
$$= R\frac{d\varphi}{dx}$$



$$\gamma_\rho = \rho\frac{d\varphi}{dx}$$

距圆心为ρ任一点处的γ_ρ与到圆心的距离ρ成**正比**。

二、 物理关系(Physical Relationship)



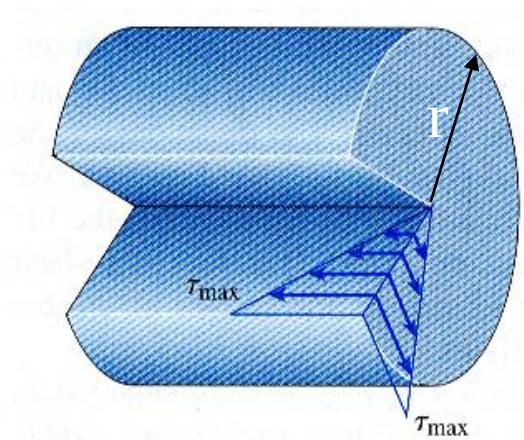
由剪切胡克定律

$$\tau = G\gamma$$

$$\tau_\rho = G\gamma_\rho = G\rho\frac{\mathrm{d}\varphi}{\mathrm{d}x}$$

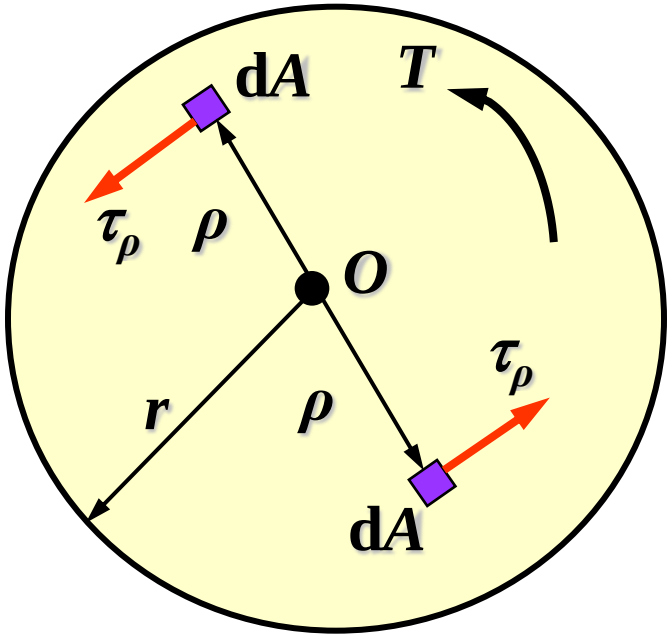
同一圆周上各点切应力τ_ρ均相同，且其值与ρ成正比，τ_ρ与半径垂直。

$$\tau_{\max} = ?$$



三、静力关系 (Static Relationship)

1、公式的建立(Establish the formula)



$$\left.\begin{aligned}&\int_A \rho\tau_\rho \mathrm{d}A = T\\&\tau_\rho = G\rho\frac{\mathrm{d}\varphi}{\mathrm{d}x}\end{aligned}\right\}\quad \int_A \rho\cdot G\cdot\rho\cdot\frac{\mathrm{d}\varphi}{\mathrm{d}x}\cdot\mathrm{d}A = T$$

$$\left.\begin{aligned}&G\frac{\mathrm{d}\varphi}{\mathrm{d}x}\int_A \rho^2\mathrm{d}A = T\\&\int_A \boldsymbol{\rho}^2\mathbf{d}\boldsymbol{A} = \boldsymbol{I}_\mathbf{p}\end{aligned}\right\}\quad \boxed{\frac{\mathrm{d}\varphi}{\mathrm{d}x} = \frac{T}{GI_\mathrm{p}}}$$

$$\left.\begin{aligned}&\boxed{\frac{\mathrm{d}\varphi}{\mathrm{d}x} = \frac{T}{GI_\mathrm{p}}}\\&\tau_\rho = G\rho\frac{\mathrm{d}\varphi}{\mathrm{d}x}\end{aligned}\right\}\quad \boxed{\boldsymbol{\tau}_\rho = \frac{\boldsymbol{T\rho}}{\boldsymbol{I}_\mathbf{P}}}$$

ρ — 求应力的点到圆心的距离。

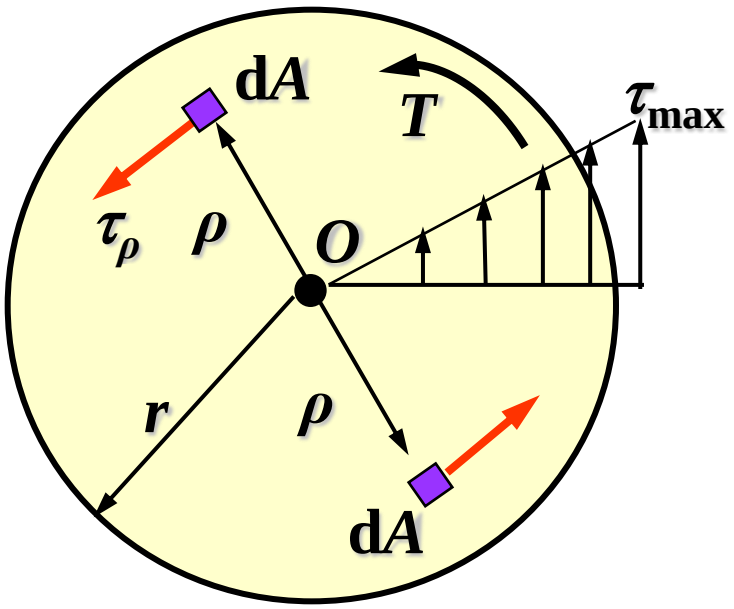
I_p —横截面对圆心的极惯性矩。纯几何量，无物理意义。

单位：mm^4，m^4。

2、τ_{max} 的计算

$$\tau_{max}=\frac{T\rho_{max}}{I_p}=\frac{T}{\dfrac{I_p}{\rho_{max}}}=\frac{T}{W_t}$$

$$W_t=\frac{I_p}{\rho_{max}}$$



W_t 称作抗扭截面系数，单位为 mm^3 或 m^3。

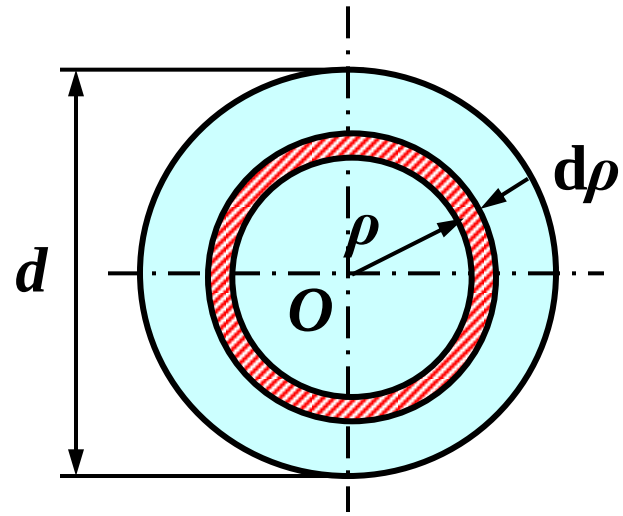
$$\boxed{\tau_{max}=\frac{T}{W_t}}$$

仅适用于各向同性、线弹性材料，在小变形时的等圆截面直杆。

3、极惯性矩和抗扭截面系数的计算 (calculating the polar moment of inertia &section modulus under torsion)

（1）实心圆截面

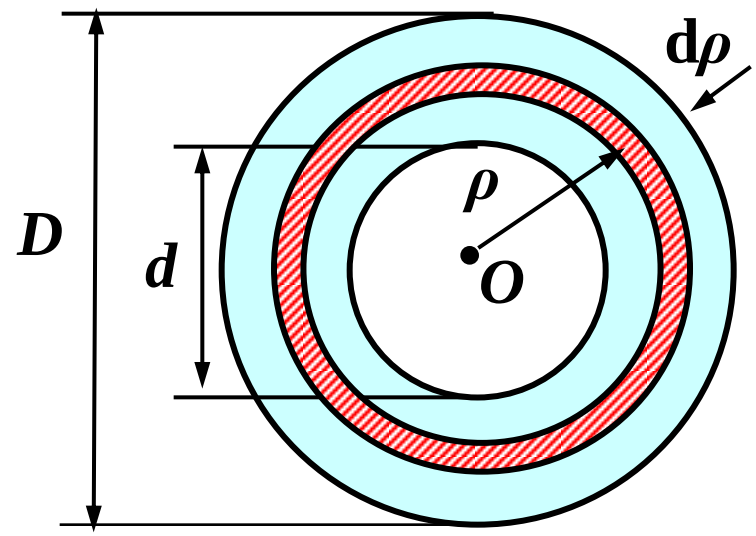
$$\mathbf{d}A = 2\pi\rho(\mathbf{d}\rho)$$



$$I_p = \int_A \rho^2 \mathrm{d}A = \int_0^{\frac{d}{2}} 2\pi\rho^3 \mathrm{d}\rho = \frac{\pi d^4}{32}$$

$$W_t = \frac{I_p}{\rho_{max}} = \frac{\pi d^4/32}{d/2} = \frac{\pi d^3}{16}$$

（2）空心圆截面



$$I_p = \frac{\pi D^4(1-\alpha^4)}{32}$$

其中 $\alpha = \dfrac{d}{D}$

$$W_t = \frac{\pi D^3}{16}(1-\alpha^4)$$

四、强度条件 (Strength Condition)

1、 数学表达式(Mathematical formula)

$$\boxed{\tau_{\max} = \frac{T_{\max}}{W_t} \le [\tau]}$$

2.强度条件的应用(Application of strength condition)

$$\tau_{\max} = \frac{T_{\max}}{W_t} \le [\tau] \begin{cases} \text{强度校核} & \dfrac{T_{\max}}{W_t} \le [\tau] \\ \text{设计截面} & W_t \ge \dfrac{T_{\max}}{[\tau]} \\ \text{确定许可载荷} & T_{\max} \le W_t[\tau] \end{cases}$$

§ 3-5 杆在扭转时的变形

(Torsional deformation of circular bars)

1、扭转角

圆轴扭转时的变形是用扭转角φ来度量。

$$\frac{\mathrm{d}\varphi}{\mathrm{d}x}=\frac{T}{GI_{\mathrm{p}}}$$

其中dφ代表相距为dx的两横截面间的相对扭转角。

$$\varphi=\int_{l}\mathrm{d}\varphi=\int_{0}^{l}\frac{\mathrm{T}}{\mathrm{GI}_{\rho}}\mathrm{dx}$$

$\boldsymbol{GI_{\mathrm{p}}}$ 称作抗扭刚度。

扭转角
- 等扭矩、等截面 $\boxed{\varphi=\dfrac{Tl}{GI_{p}}}$
- 阶梯扭矩、阶梯截面 $\boxed{\varphi=\sum_{i=1}^{l}\dfrac{T_{i}l_{i}}{GI_{p_{i}}}}$

2、单位长度扭转角(Angle of twist per unit length)

$$\varphi' = \frac{T}{GI_\rho}$$ 单位：rad/m

3、刚度条件（Stiffness condition)

$$\boldsymbol{\varphi'_{max} \le [\varphi']}$$

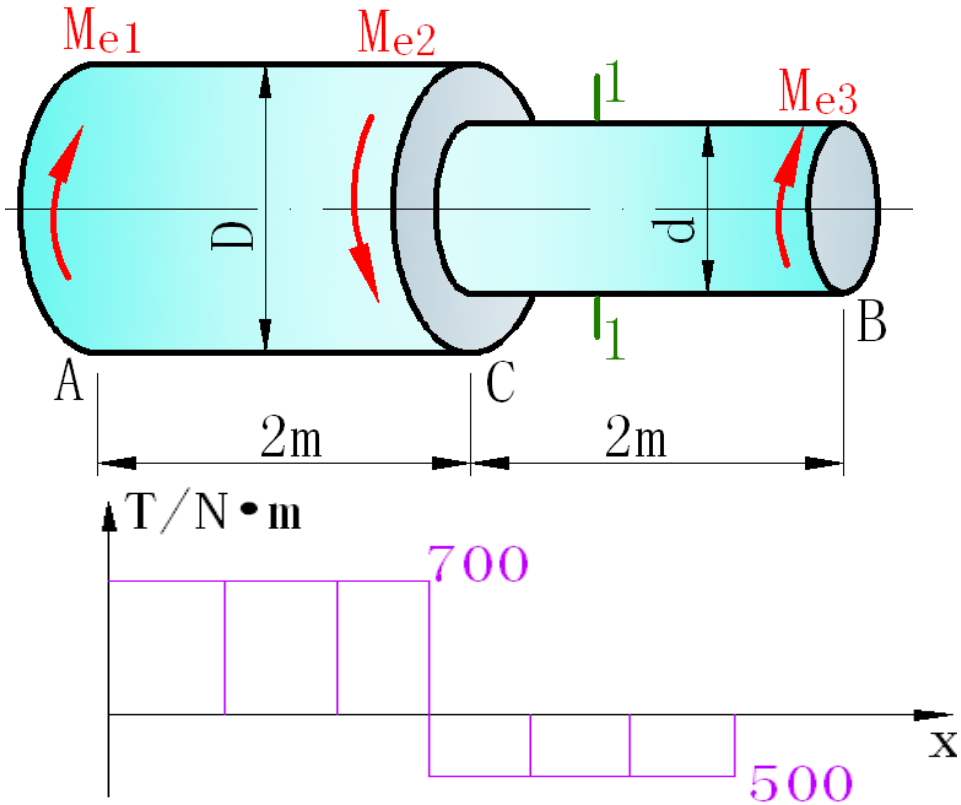
$\boldsymbol{[\varphi']}$称作许可单位长度扭转角

$$\boldsymbol{\varphi'_{max} = \frac{T_{max}}{GI_p} \le [\varphi'] \quad (rad/m)}$$

$$\varphi'_{max} = \frac{T}{GI_p} \cdot \frac{180}{\pi} \le [\theta] \quad (°/m)$$

例题3-3 阶梯轴，D=50mm，d=40mm， M_{e1}=700N m，M_{e2}=1200N m，M_{e3}=500N m，[τ] = 100MPa，切变模量为 G = 80GPa，轴的许可扭角[φ′] = 2°/m。

求：（1）1-1截面上ρ=10mm处的应力；（2）校核强度；（3）校核刚度；（4）计算各段扭转角。



$$\tau_{1\text{-}1\rho} = 19.9\text{MPa}$$

$$\tau_{max} = 39.8\text{Mpa}$$

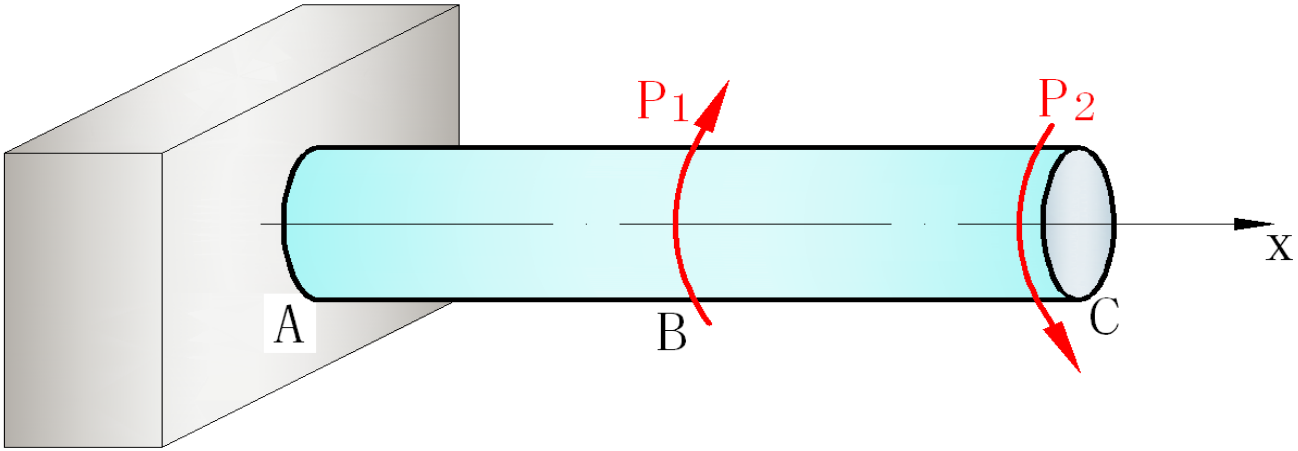
$$\varphi'_{max} = 1.42\ ^{o}/\text{m}$$

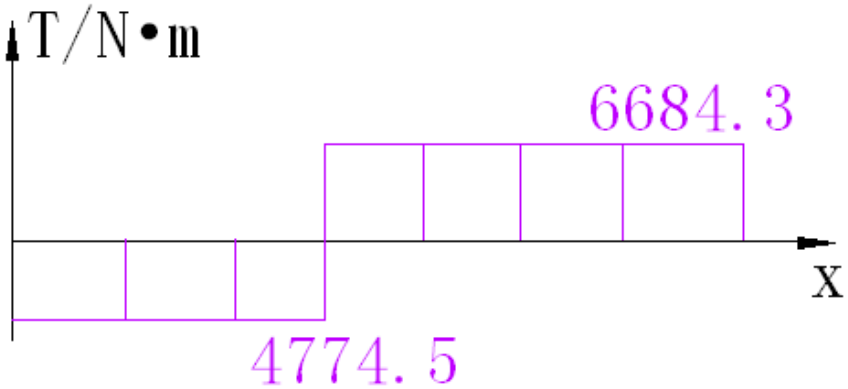
$$\varphi_{AC} = 1.64^{o}$$

$$\varphi_{CB} = -2.84^{o}$$

$$\varphi_{AC} = -1.2^{o}$$

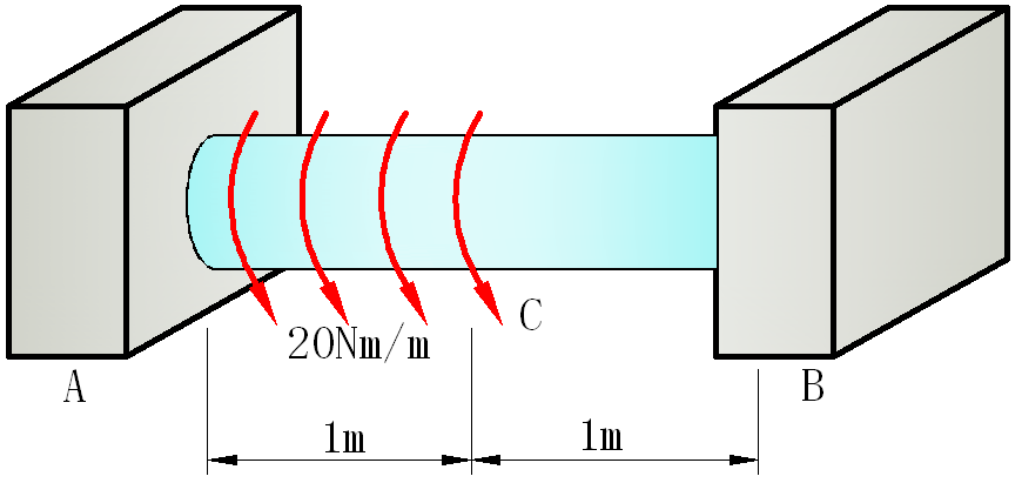
例题3-4　图示圆轴, P_1=360kW，P_2=210kW，n=300r/min，[τ] = 40MPa，切变模量为G = 80GPa，轴的许可扭角[φ′] = 0.005236rad/m。设计轴径。

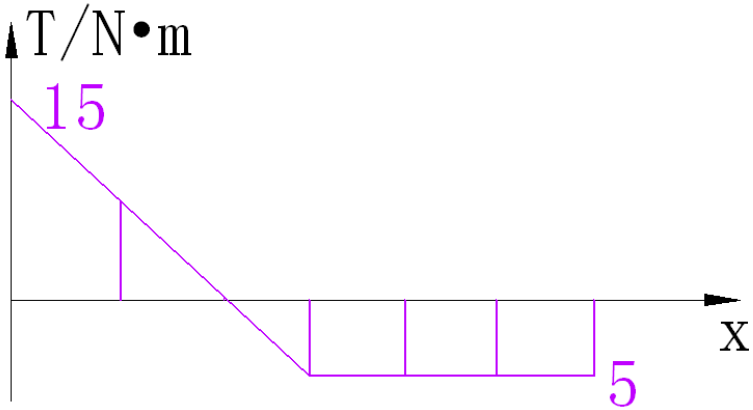




d≥0.1129m

例题3-4　两端固定圆轴AB，AC段受均布、集度m=20N m/m的力偶作用。求固定端的约束力偶，并作出扭矩图。

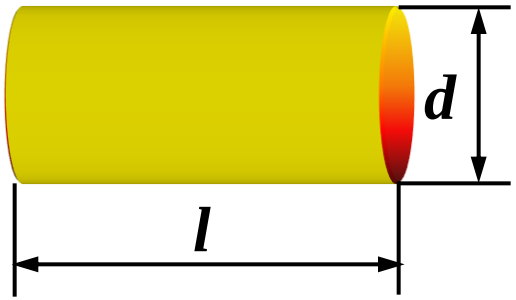




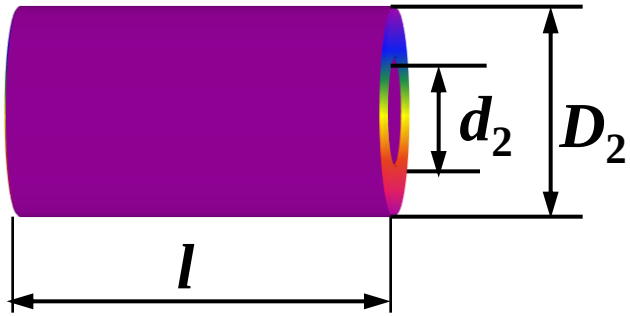
m_A=15N m

m_B=5N m

例题3-5 实心圆轴1和空心圆轴2（图a、b）材料、长度、受到的载荷均相等，强度一样。若空心圆轴的内外径之比 $\alpha = 0.8$，试求空心圆截面的外径和实心圆截面直径之比及两轴的重量比。



(a)



(b)

$$\tau_{\max 1} = \tau_{\max 2}$$

$$\tau_{\max 1} = \frac{T}{W_{t1}} \qquad \tau_{\max 2} = \frac{T}{W_{t2}}$$

$$\frac{T}{W_{t1}} = \frac{T}{W_{t2}}$$

$$W_{t1} = W_{t2} = \frac{\pi d^3}{16} = \frac{\pi D_2^3 (1-\alpha^4)}{16}$$

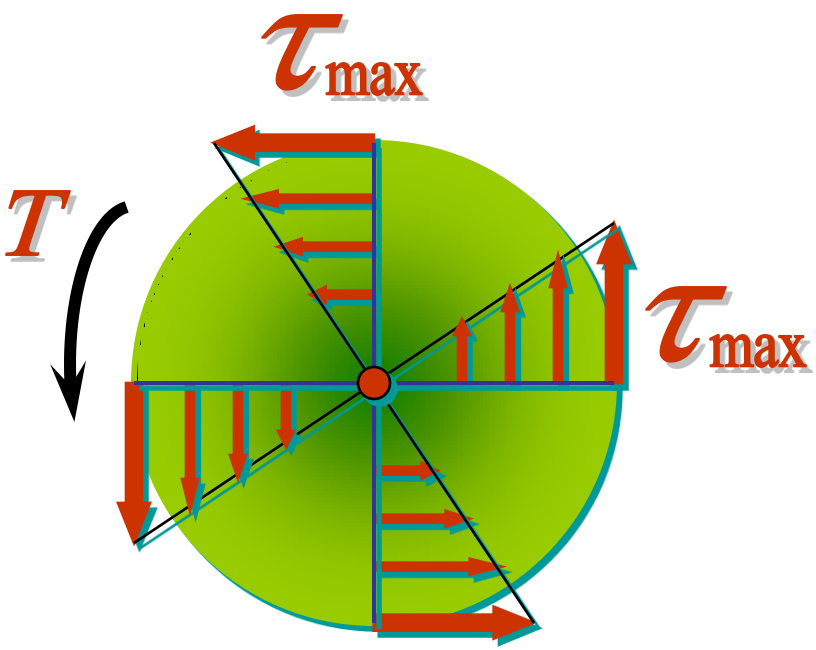
$$\frac{\pi d^3}{16} = \frac{\pi D_2^3 (1-\alpha^4)}{16}$$

$$\frac{D_2}{d} = \sqrt[3]{\frac{1}{1-0.8^4}} = 1.194$$

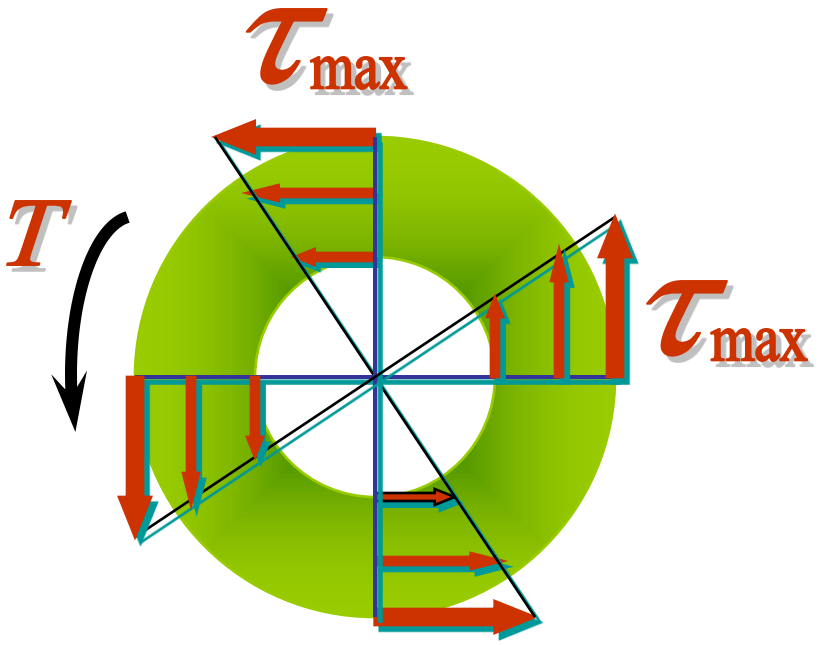
两轴材料、长度均相同，故两轴的重量比等于两轴的横截面面积之比。

$$\frac{A_2}{A_1} = \frac{\frac{\pi}{4}(D_2^2 - d_2^2)}{\frac{\pi}{4} d^2} = \frac{D_2^2 (1-\alpha^2)}{d^2} = 1.194^2 (1-0.8^2) = 0.512$$

应力分布



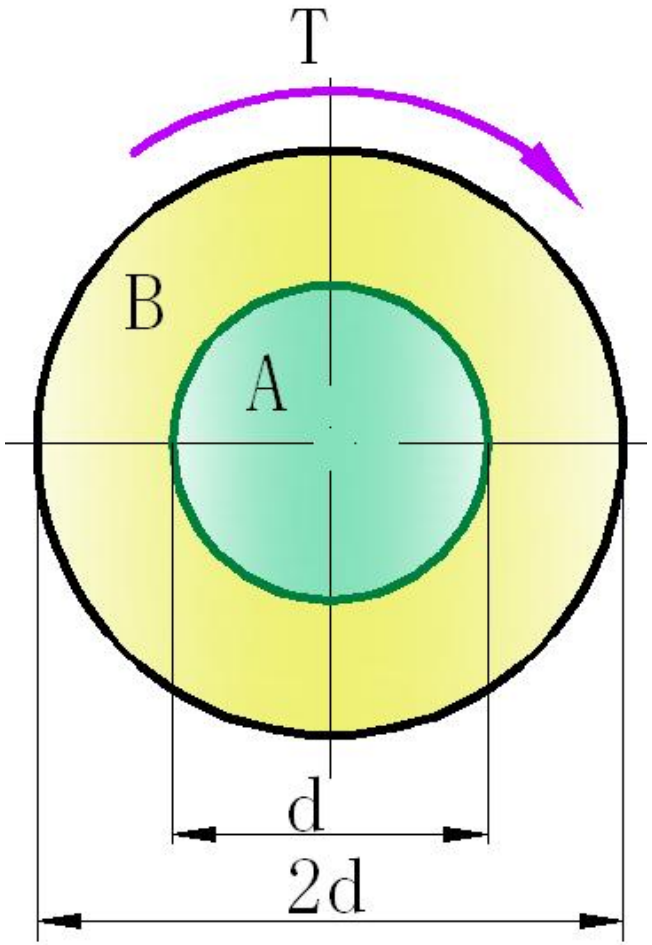
实心截面

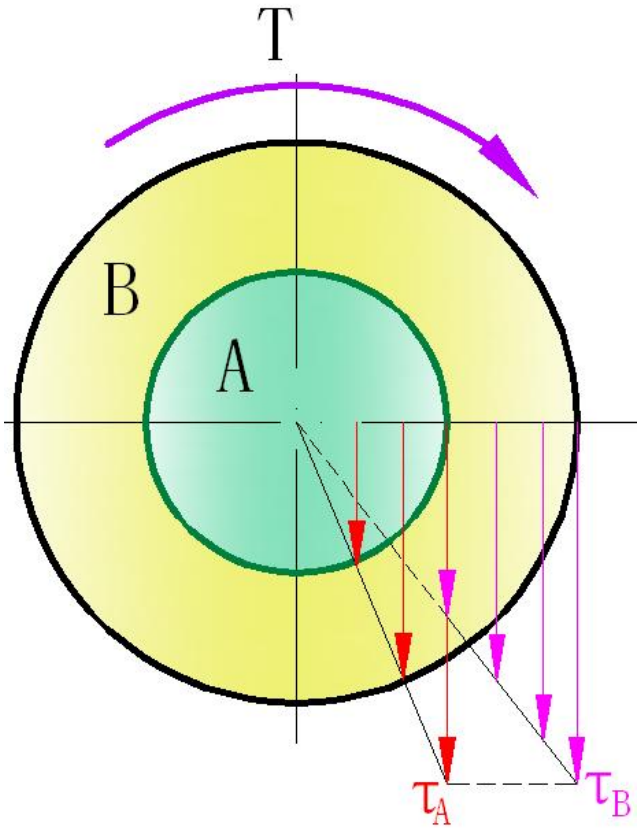


空心截面

工程上采用空心截面构件：提高强度，节约材料，重量轻，结构轻便，应用广泛。

例题3-6　受扭圆轴由实心圆轴A和空心圆轴B牢固连接而成。两轴材料的切变模量为G_A、G_B，且$G_A=2G_B$。横截面上切应力的分布规律和各轴分得的扭矩。





$$T_A=\frac{2}{17}T$$

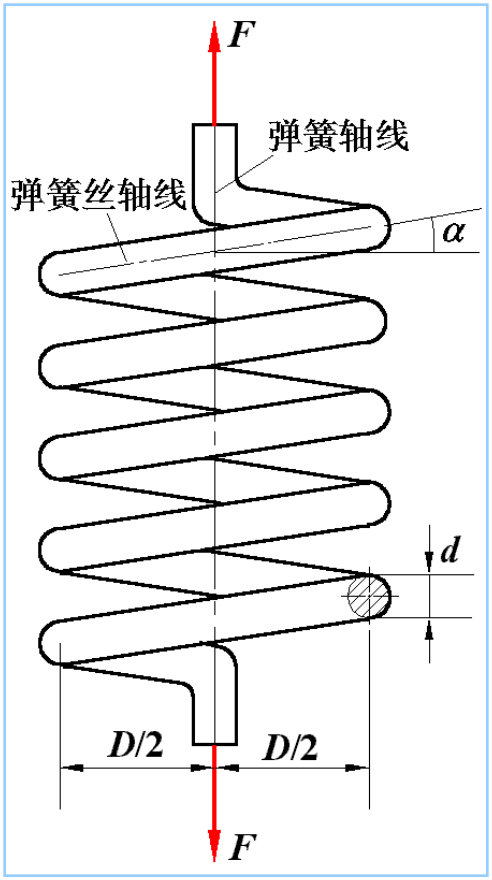
$$T_B=\frac{15}{17}T$$

§ 3-6 密圈螺旋弹簧的应力和变形

(Calculation of the stress and deformation in close-coiled helical springs)

弹簧的作用：减震、缓冲、施加载荷、储存能量。

密圈螺旋弹簧：弹簧的螺旋角 $\alpha<5°$，且$D>>d$，绕在圆柱面上。



一、弹簧丝横截面上的应力

1、假设

（1）因为$D>>d$，所以弹簧丝曲率影响忽略不计，可用直杆计算。

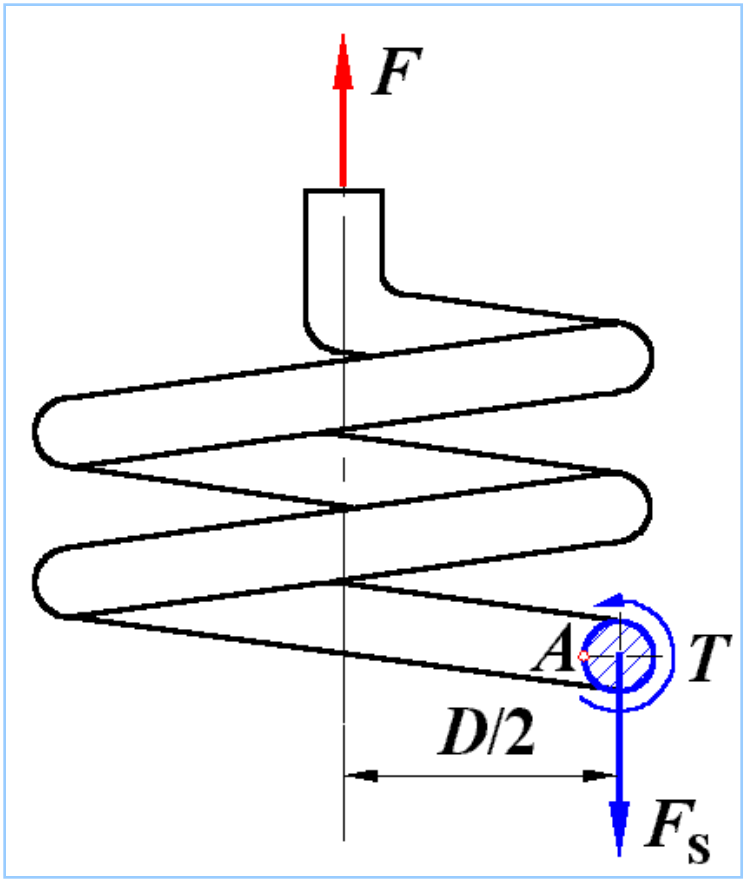
（2）因为α<5°，所以弹簧丝横截面与弹簧轴线在同一平面内；

2、内力的计算(Calculation of internal force)

方法：截面法

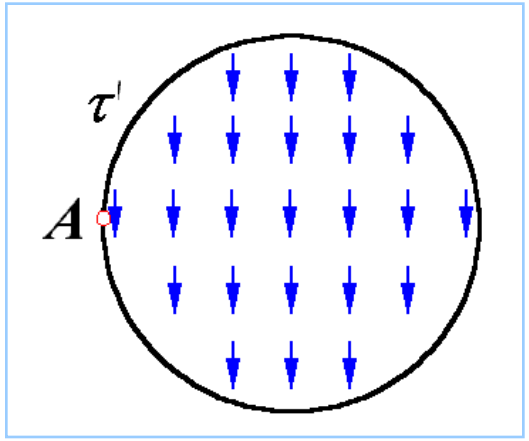
簧丝的横截面上有两个内力分量。

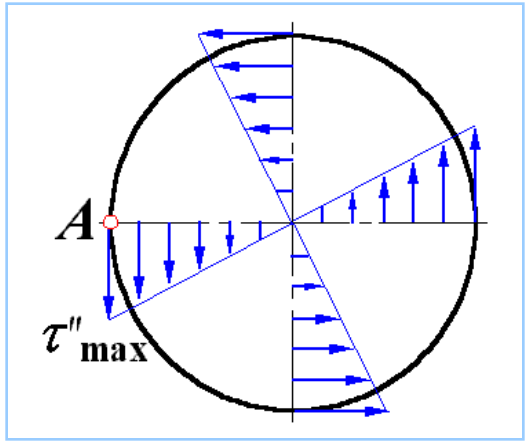
$$F_s = F \quad T = \frac{FD}{2}$$

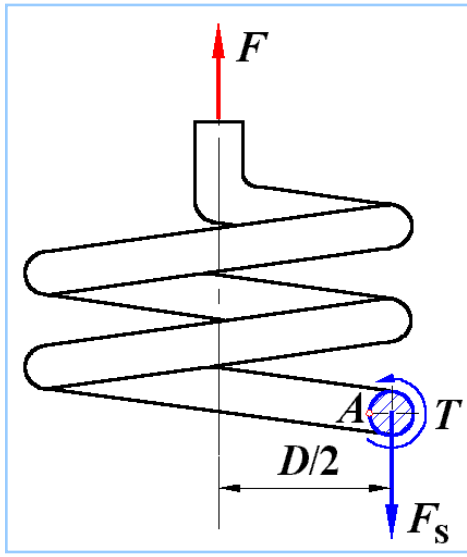


变形形式：剪切和扭转组合变形。

3、应力的计算(Calculation of stress)







$$\tau' = \frac{4F_s}{\pi d^2} = \frac{4F}{\pi d^2}$$

$$\tau''_{max} = \frac{T}{W_p} = \frac{FD}{2}\frac{16}{\pi d^3} = \frac{8FD}{\pi d^3}$$

$$\tau_{max} = \tau''_{max} + \tau' = \frac{8FD}{\pi d^3}\left(1 + \frac{d}{2D}\right)$$

公式修正的原因:

（1）当D/d较小，会引起很大的误差;

（2）假定剪切引起的切应力是均匀分布的。

$$\tau_{\max}=\left(\frac{4c-1}{4c-4}+\frac{0.615}{c}\right)\frac{8FD}{\pi d^3}=k\frac{8FD}{\pi d^3}$$

式中 $c=\dfrac{D}{d}$, $k=\dfrac{4c-1}{4c-4}+\dfrac{0.615}{c}$

c为弹簧指数。k为曲度系数，可查教材中的表3.1。

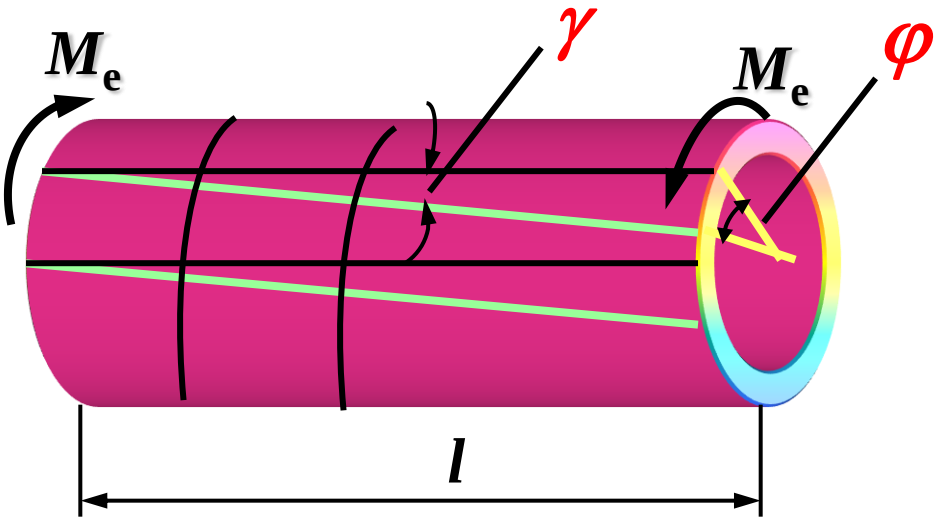
4、强度条件(Strength condition)

$$\tau_{\max}\le[\tau]$$ $[\tau]$－纯剪切许用切应力

二、弹簧的变形(Deformation of the spring)

1、应变能的计算
(Calculation of strain energy)

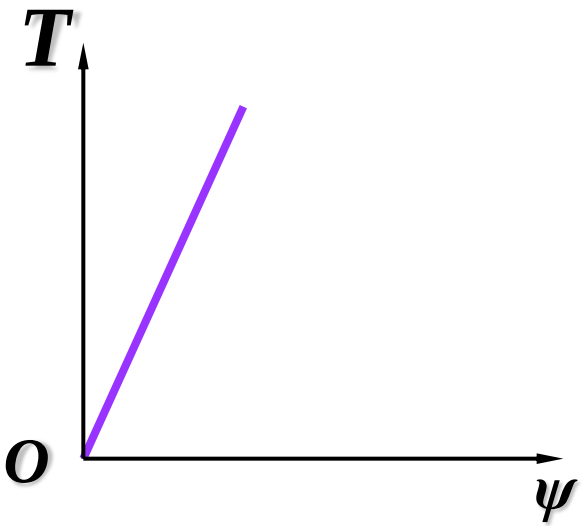
若只考虑簧杆扭转的影响。

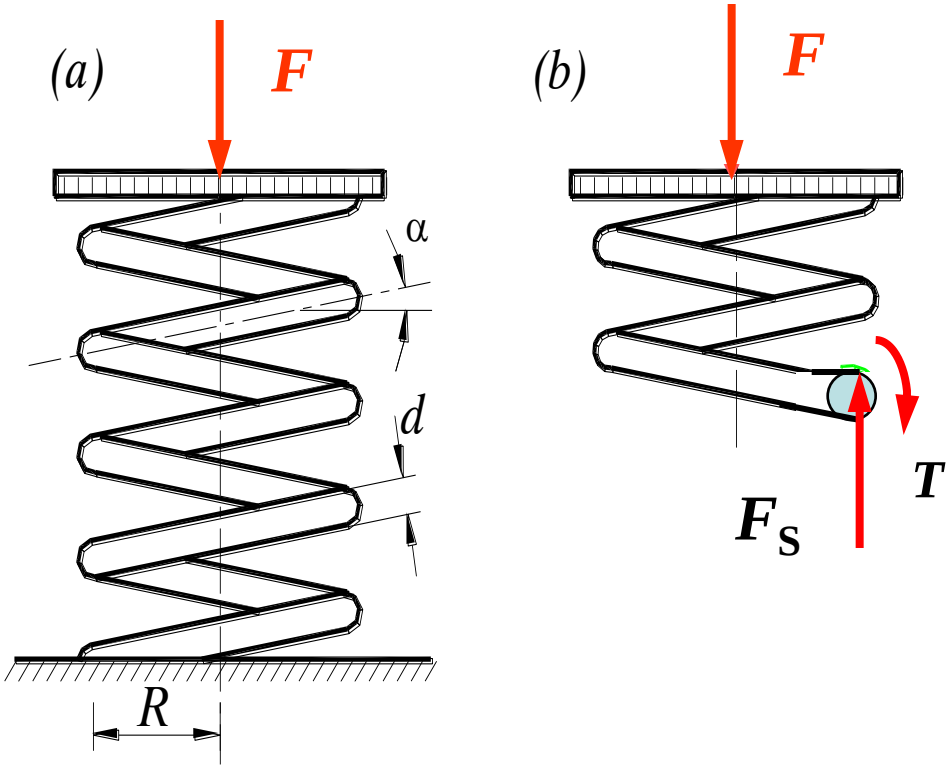


$$W = V_\varepsilon$$

$$W = \frac{1}{2} M_e \varphi$$

$$V_\varepsilon = \frac{1}{2} M_\varepsilon \varphi = \frac{1}{2} \cdot T \cdot \frac{TL}{GI_\rho} = \frac{T^2 L}{2GI_\rho}$$



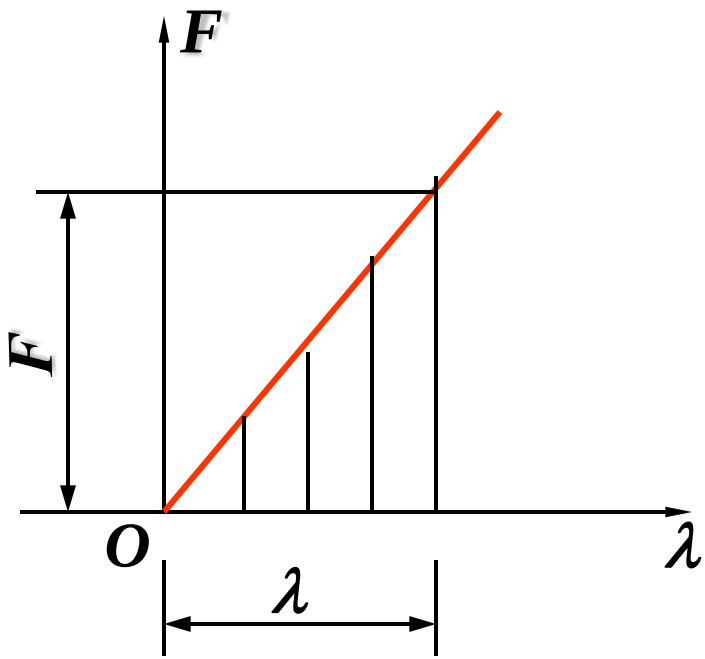


$$\upsilon_\varepsilon = \frac{T^2 l}{2GI_p} \xrightarrow[l = n\pi D]{T = \frac{FD}{2}} \upsilon_\varepsilon = \frac{4F^2 D^3 n}{Gd^4}$$

2、外力做的功(Work of the external force)

当弹簧的变形为 l 时，外力所做的功为

$$W = \frac{1}{2}F\lambda$$



3、功能原理(Work-energy principle)

$$V_\varepsilon = W$$

$$\frac{1}{2}F\lambda = \frac{4F^2D^3n}{Gd^4}$$

$$\lambda = \frac{8FD^3n}{Gd^4} = \frac{64FR^3n}{Gd^4} \quad 令 c = \frac{Gd^4}{8D^3n} = \frac{Gd^4}{64R^3n}$$

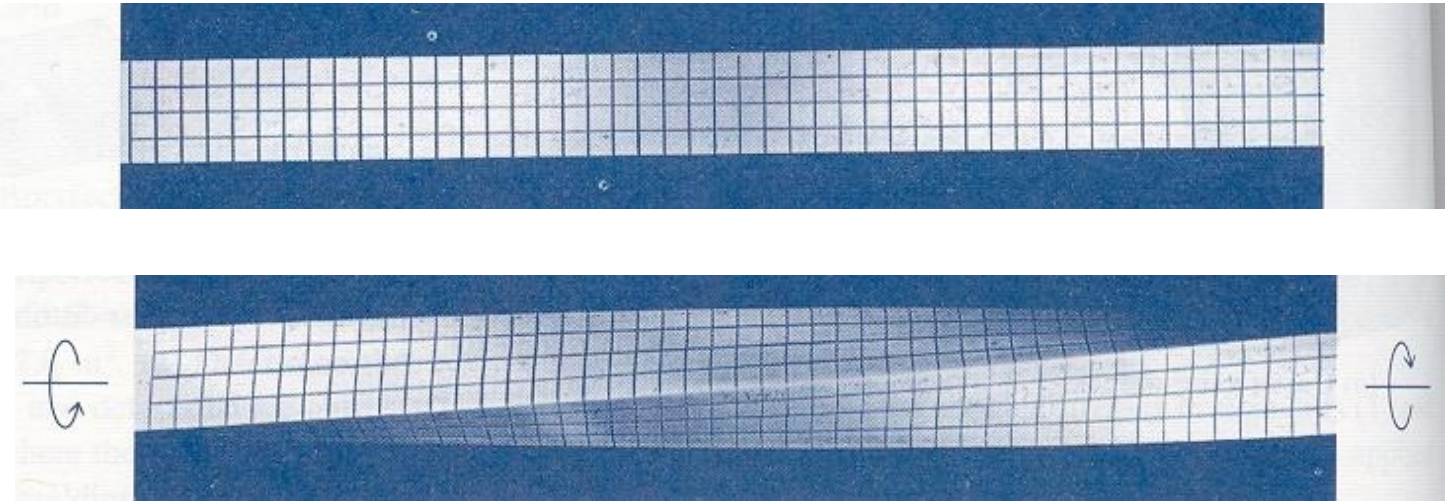
$$得 \lambda = \frac{F}{c}$$

c —弹簧刚度

§ 3-7 非圆截面杆的扭转

(Torsion of noncircular prismatic bars)

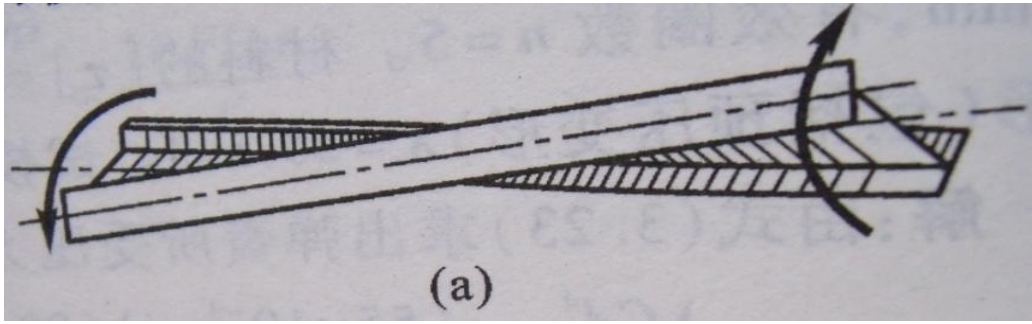
一、基本概念(Basic concepts)

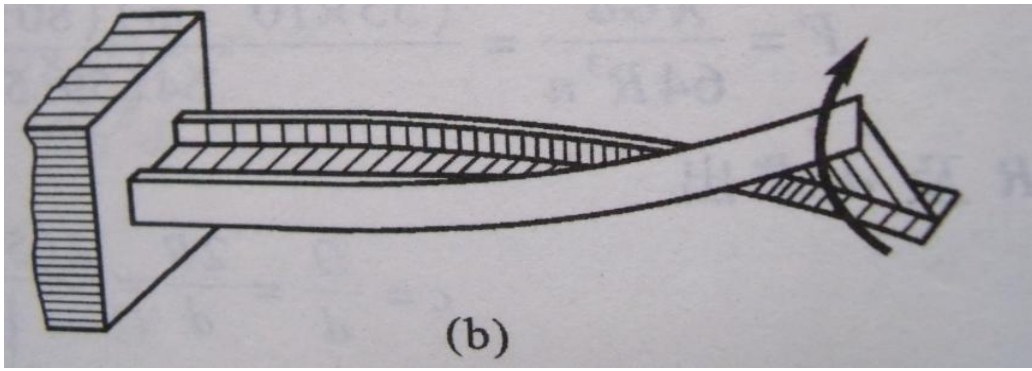


1、翘曲(warping)：非圆截面等直杆扭转平面假设不成立。即各截面发生，不再保持平面。由等直圆杆扭转时推出的应力、变形公式不适用，须由弹性力学方法求解。

2、自由扭转：当等直杆在两端受外力偶作用，横截面的翘曲不受限制，端面可以自由翘曲时，任意两相邻截面的翘曲程度完全相同。

自由扭转时横截面上仍然只有切应力而没有正应力。

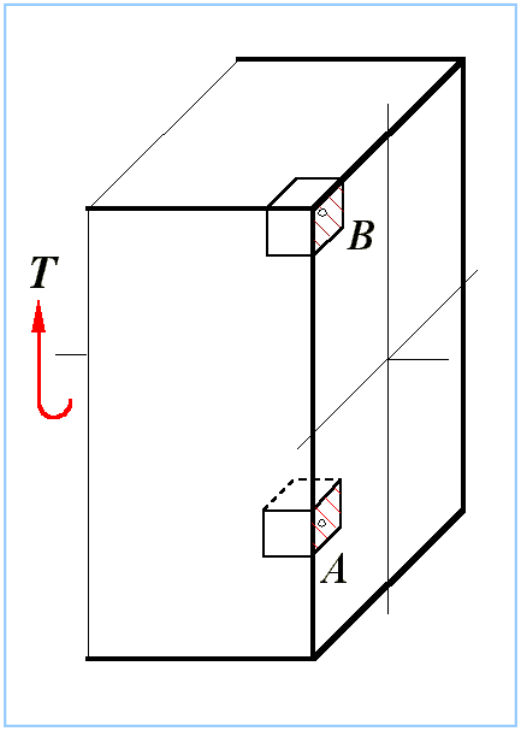


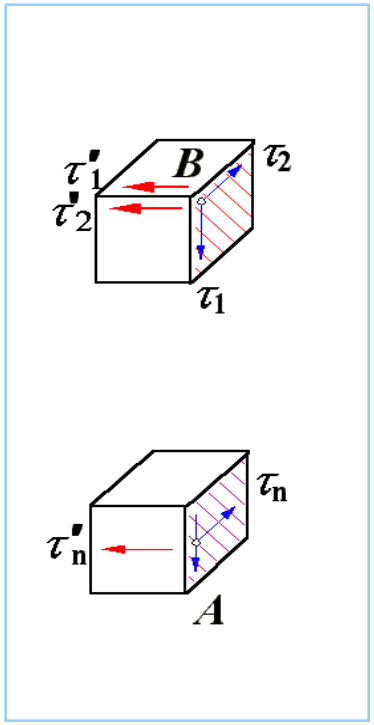


3、约束扭转：若杆的两端受到约束而不能自由翘曲，则相邻两横截面的翘曲程度不同。

约束扭转时这将在横截面上引起附加的正应力。

4、切应力分布特点





$$\tau_1' = \tau_2' = 0$$

故 $\tau_1 = \tau_2 = 0$

$$\tau_n' = 0$$

故 $\tau_n = 0$

（1）横截面上角点处，切应力为零。

（2）横截面边缘各点处，切应力 // 截面周边。

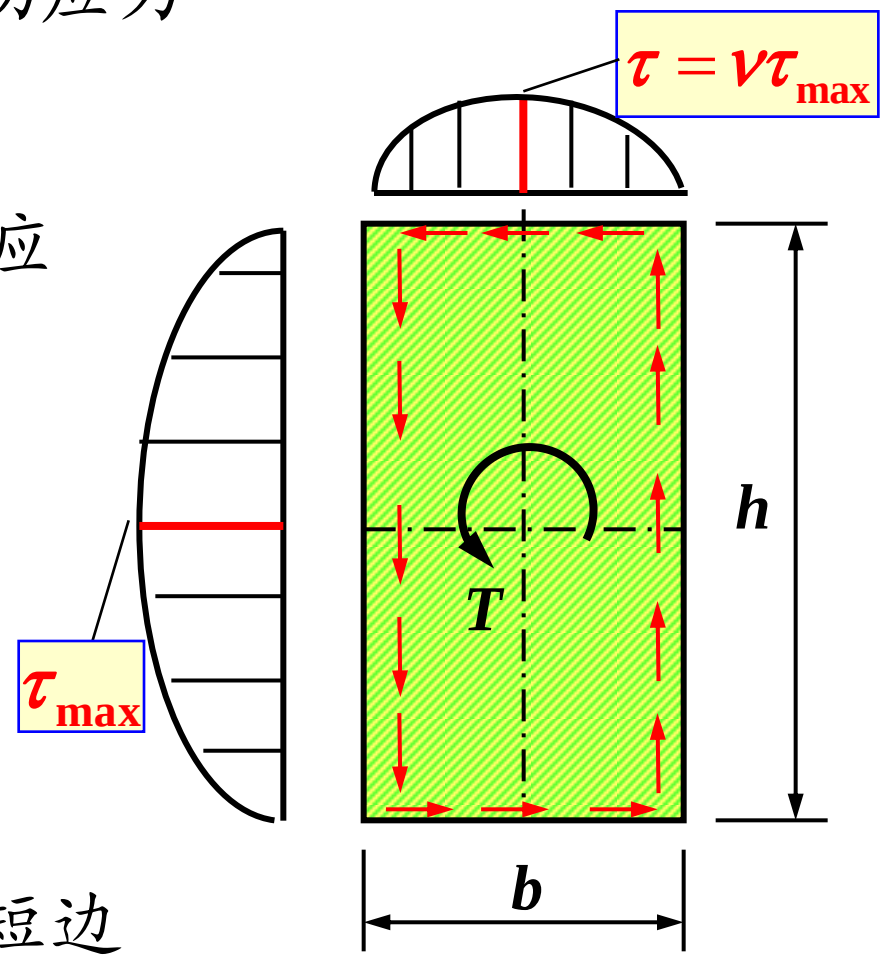
二、矩形截面(Rectangular cross section)

矩形截面扭转时，横截面切应力如图所示，边缘上各点的切应力形成与边界相切的顺流。

整个横截面上的最大切应力发生在长边的中点。

$$\tau_{max} = \frac{T}{W_t} \qquad W_t = \alpha h b^2$$

$$\varphi = \frac{Tl}{GI_t} \qquad I_t = \beta h b^3$$



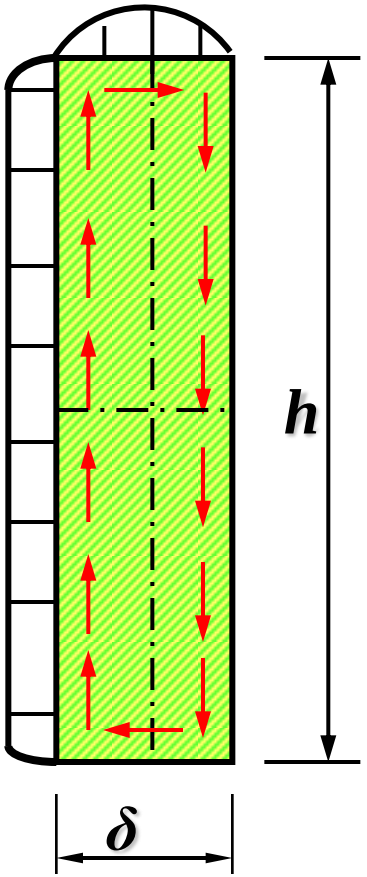
短边中点的切应力τ是短边上的最大切应力,且$\tau = \nu\tau_{max}$ 。

三、狭长矩形（Long narrow rectangle)

切应力在沿长边各点处的方向均与长边相切其数值除在靠近顶点处以外均相等。

狭长矩形截面的 I_t 和 W_t

$$I_t = \frac{1}{3} h \delta^3 \qquad W_t = \frac{1}{3} h \delta^2$$



第三章结束